

Relational Multiplicity and the Emergence of Internal Gauge Structure in Motion–Staticity Quantum Comparison

Mingyang Han

School of Mathematical Sciences, Shanghai Jiao Tong University

Shanghai, 200240, China

hanmingyang@sjtu.edu.cn

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Abstract The motion–staticity (MS) reconstruction programme begins from operational generator–displacement response and develops spacetime geometry before a gravitational or quantum dynamics is imposed. Previous extensions of the programme have identified two distinct routes beyond the classical response algebra: canonical coherent deformations of the soldering variables and projective inter-event comparison. In particular, an enlarged relative-comparison sector can, under additional dynamical assumptions, support projective half-integer boundary excitations without introducing microscopic spinor variables. The existence of such projective carriers, however, does not by itself explain the origin of internal quantum degrees of freedom or internal gauge structure.

In this paper we study the many-body composition of minimal projective comparison carriers, taking a spin- $\frac{1}{2}$ infrared sector as the starting point. For N such carriers the collective geometric action admits the decomposition

$$\mathcal{H}_N = \left(V_{\frac{1}{2}}\right)^{\otimes N} = \bigoplus_j V_j \otimes M_j,$$

where the factors V_j carry the collective rotational representation while the multiplicity spaces M_j are invisible to that geometric action. We show that these multiplicity spaces are not merely degeneracy labels. Pairwise neutral-comparison projectors act noncommutatively on them and, already in the minimal three- and four-carrier examples, generate a complete qubit operator algebra. In total-spin-neutral networks every individual carrier is maximally mixed, while the invariant relational state space retains an asymptotically extensive information capacity. This gives a precise sense in which quantum information can reside entirely in relations while remaining inaccessible to one-body geometric observables.

We then promote protected multiplicity subspaces to event-dependent bundles. Their moving embeddings into a larger comparison Hilbert space define Grassmannian-valued fields and induce matrix-valued comparison connections. A fixed multiplicity space with only a changing basis produces zero curvature; nontrivial internal curvature requires the physical low-energy projector itself to vary. The maximal commutant group $U(m)$ is therefore distinguished from the dynamically realized holonomy group. Finally, by requiring comparison transport to preserve fusion intertwiners among relational sectors, we obtain a route from sector composition to a common tensor-automorphism group acting in different representations on different matter sectors. This provides a reconstruction mechanism for internal gauge structure from relational multiplicity and fusion, without identifying the commutant with a gauge group by assumption. The appearance of an independently propagating Yang–Mills field remains a separate infrared dynamical requirement.

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1 Introduction

A central difficulty in attempts to unify quantum theory with spacetime geometry is that the two structures are usually introduced at logically different levels. Spacetime geometry is specified through a metric, tetrad, or connection, while quantum theory is introduced through a Hilbert space, operator algebra, or path integral. The subsequent problem is then formulated as one of quantizing an already defined geometry, or of embedding gravity into an already defined quantum framework. The motion–staticity (MS) programme takes a different route. Its starting point is an operational question: which responses can be assigned to a generator when it is compared with a local displacement? Under suitable composition, continuity, and faithfulness conditions, this response structure determines a soldering map and therefore a local Lorentzian metric before the gravitational constitutive law is fixed [?, ?]. The resulting separation between operational geometry and dynamical response has subsequently been used to distinguish the reconstruction of spacetime from the independent problem of determining gravitational stiffness.

The same separation becomes important when quantum structure is introduced. An exact unitary promotion of the local generator–displacement response does not, by itself, produce a noncommutative quantum geometry. Under exact additivity in the relevant response labels, the local response algebra remains commutative and admits an ordinary spectral description [?]. A genuine quantum

completion therefore requires additional structure outside the exact response algebra. Two such structures have been investigated in the MS setting. Coherent deformations of the soldering configuration space give a canonical cotangent completion with a Heisenberg–Weyl algebra, while finite rotational comparison can be represented projectively because the quantum state is a ray rather than a vector [?]. The latter construction distinguishes the continuous Weyl phase from the global spinorial class of rotational comparison. For orthogonal π rotations one may define a rephasing-invariant quantity

$$\Xi = U_x(\pi)U_y(\pi)U_x(\pi)^{-1}U_y(\pi)^{-1},$$

whose value $\Xi = -I$ detects the nontrivial projective rotational class. The nontrivial center of the spin cover acts trivially on vector and tensor geometric observables, placing projective spinorial sectors naturally outside the algebra generated by purely tensorial geometry.

A separate relative-comparison construction has provided a conditional dynamical mechanism by which such a projective sector may arise without postulating microscopic spinor variables. An independent quantum comparison transport, distinct from the geometric Lorentz connection, possesses a low-energy rotational triplet under the compact little group. When these bosonic $j = 1$ comparison modes are bound to occupied string segments, their nearest-neighbour Hilbert space decomposes as

$$1 \otimes 1 = 0 \oplus 1 \oplus 2.$$

If the symmetric-traceless $j = 2$ channel is sufficiently penalized, the effective comparison chain can enter a Haldane-type phase whose open endpoints carry the nontrivial projective class $\Xi = -I$ [?]. This mechanism is deliberately conditional. The required channel ordering, deconfinement, relativistic localization, and spin-statistics relation are not consequences of the original local response axioms. Nevertheless, it provides a concrete starting point for a question that is logically subsequent to spin fractionalization and is the subject of the present paper: once projective comparison carriers exist, can their many-body composition generate internal quantum structure that is invisible to ordinary geometry?

We address this question by taking a minimal spin- $\frac{1}{2}$ projective carrier as an infrared input and studying its tensor powers. Let

$$V_{\frac{1}{2}} \simeq \mathbb{C}^2$$

denote the minimal carrier and consider

$$\mathcal{H}_N = \left(V_{\frac{1}{2}}\right)^{\otimes N}.$$

Under the collective $SU(2)$ action this space decomposes as

$$\mathcal{H}_N = \bigoplus_j V_j \otimes M_j.$$

The factors V_j carry the irreducible collective rotational representation, whereas the multiplicity spaces M_j record the different ways in which the same geometric representation occurs. Algebraically this is standard representation theory. The point of the MS interpretation is different: once the collective rotational action is identified with the geometric comparison action, the multiplicity spaces are precisely those coherent degrees of freedom that remain after the geometric representation has been fixed. They are therefore natural candidates for internal relational state spaces.

The first nontrivial examples already exhibit the relevant structure. For three carriers,

$$\left(V_{\frac{1}{2}}\right)^{\otimes 3} \simeq V_{\frac{3}{2}} \oplus \left(V_{\frac{1}{2}} \otimes \mathbb{C}^2\right),$$

so that a geometric spin- $\frac{1}{2}$ carrier occurs together with a two-dimensional multiplicity space. For four carriers,

$$\left(V_{\frac{1}{2}}\right)^{\otimes 4} \simeq V_2 \oplus \left(V_1 \otimes \mathbb{C}^3\right) \oplus \left(V_0 \otimes \mathbb{C}^2\right).$$

In particular,

$$\mathcal{I}_4 := \text{Inv}_{SU(2)} \left[\left(V_{\frac{1}{2}}\right)^{\otimes 4} \right] \simeq \mathbb{C}^2.$$

Thus a completely rotationally neutral four-carrier network already contains a genuine two-dimensional quantum state space. Collective geometry acts trivially on this space, but the state need not be trivial.

The central observation of the present work is that these multiplicity spaces support a noncommutative algebra of purely relational observables. Define the singlet projector for a pair of minimal carriers by

$$P_{ij}^{(0)} = \frac{1}{4} (I - \sigma_i \cdot \sigma_j).$$

Each $P_{ij}^{(0)}$ is a rotational scalar and therefore commutes with the collective geometric action. It asks a relational question: whether the carriers i and j occupy their neutral spin channel. Different such questions, however, need not be compatible. In the two-dimensional multiplicity sector one may choose a basis in which

$$P_{12}^{(0)} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_{23}^{(0)} = \begin{pmatrix} \frac{1}{4} & -\frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix},$$

and hence

$$\left[P_{12}^{(0)}, P_{23}^{(0)} \right] \neq 0.$$

The noncommutativity therefore survives after all collective rotational information has been removed. In the minimal examples, pairwise neutral-comparison operators generate the full matrix algebra on the multiplicity space. Internal quantum observables can consequently be realized as geometry-preserving rearrangements of relational structure rather than as additional coordinates attached to an otherwise complete geometric object. The relevant algebraic facts are closely related to the standard centralizer, Schur–Weyl, and Temperley–Lieb descriptions of spin tensor powers [?, ?, ?]; the proposed physical interpretation within the MS reconstruction is the focus here.

The same construction gives a precise form to the claim that the information is relational rather than one-body. If

$$|\Psi\rangle \in \text{Inv}_{SU(2)} \left[\left(V_{\frac{1}{2}}\right)^{\otimes 2n} \right],$$

then the reduced state of every individual carrier satisfies

$$\rho_i = \frac{1}{2} I.$$

No one-carrier tomography can therefore recover the state stored in the invariant multiplicity sector. Moreover, if P_{inv} denotes the projector onto a total-spin-zero subspace, then a rank-one geometric vector operator has no matrix element within that subspace:

$$P_{\text{inv}} J_i^a P_{\text{inv}} = 0.$$

Consequently, a one-body geometric perturbation acts on neutral relational information only through leakage to nonzero-spin sectors at first order; its effective action within the relational space begins at higher order and is expressed through rotational scalars such as $\mathbf{J}_i \cdot \mathbf{J}_j$. This gives an operational

meaning to “internal”: the information is not merely degenerate under geometry, but is invisible to one-body geometric probes while remaining accessible to multi-carrier relational measurements.

The capacity of these neutral spaces is also large enough to support a genuine internal state sector. For $N = 2n$ minimal carriers,

$$\dim \text{Inv}_{SU(2)} \left[\left(V_{\frac{1}{2}} \right)^{\otimes 2n} \right] = C_n = \frac{1}{n+1} \binom{2n}{n},$$

the n th Catalan number. Hence

$$\log_2 C_n = N - \frac{3}{2} \log_2 N + O(1).$$

Imposing complete collective rotational neutrality therefore removes only a subextensive amount of information from the original tensor-product Hilbert space. In the large-network limit the encoding rate tends to unity. A geometrically neutral composite can thus carry an extensive amount of quantum information entirely in its relational multiplicity.

The next question is whether such multiplicity spaces can acquire the local comparison structure characteristic of an internal gauge theory. Let a protected m -dimensional relational subspace vary with the spacetime event, and represent an orthonormal local frame by

$$Z(x) = (|m_1(x)\rangle, \dots, |m_m(x)\rangle), \quad Z^\dagger Z = I_m.$$

The corresponding physical projector is

$$P(x) = Z(x)Z(x)^\dagger.$$

A local change of relational basis

$$Z(x) \mapsto Z(x)U(x), \quad U(x) \in U(m),$$

does not alter $P(x)$. Given an ambient comparison derivative D , the induced connection is

$$A = Z^\dagger DZ,$$

with the usual transformation law

$$A \mapsto U^{-1}AU + U^{-1}dU.$$

This observation alone does not produce a gauge field. If the physical projector is constant and only the basis is changed, the resulting connection is pure gauge and its curvature vanishes. Nontrivial internal curvature requires the low-energy relational subspace itself to move inside the larger comparison Hilbert space.

This distinction is most naturally formulated using Grassmannian geometry. An m -dimensional protected relational sector embedded in an N -dimensional ambient comparison space defines

$$P(x) \in \text{Gr}(m, N).$$

The induced curvature can be written directly in terms of the physical projector,

$$F = P(DP \wedge DP)P,$$

up to the conventional placement of factors of i . If

$$K = (I - P)DZ$$

denotes the extrinsic variation of the relational frame, the induced connection satisfies a Gauss-type relation in which its curvature is determined by the projection of the ambient comparison curvature together with an extrinsic term quadratic in K . Thus internal holonomy may arise from the failure of a protected low-energy relation space to admit a globally constant embedding. In the stable large- N limit the spaces $\text{Gr}(m, N)$ approach the classifying space $BU(m)$, showing that sufficiently large relational Hilbert spaces have the topological capacity required for general complex rank- m bundles. This is a kinematic capacity statement, not a derivation of a particular gauge group.

A second distinction is therefore essential. The unitary group $U(m)$ of a multiplicity space is the maximal kinematic basis group, whereas the physical comparison group is determined by the holonomy of the actual induced connection:

$$G_{\text{hol}} = \overline{\text{Hol}(\nabla_M)} \subseteq U(m).$$

Its connected Lie algebra is generated by the curvature values and their parallel transports. A two-dimensional multiplicity space need not therefore produce an $SU(2)$ interaction. If all traceless curvature components are confined to a single commuting direction, the connected holonomy is Abelian. Only when the curvature explores noncommuting directions does its Lie closure become non-Abelian. This provides a concrete dynamical criterion that replaces the naive identification of a multiplicity dimension with a gauge group.

Even a non-Abelian holonomy associated with one relational sector is not yet a universal internal interaction. A physical gauge field must act consistently across different matter sectors. We propose that the relevant consistency condition is compatibility between comparison transport and fusion. Suppose relational sectors r, s, t admit an intertwiner

$$C_{rs}^t(x) : M_r(x) \otimes M_s(x) \longrightarrow M_t(x).$$

If transport from x to y is to describe the same physical comparison whether fusion is performed before or after transport, then

$$U_\gamma^{(t)} C_{rs}^t(x) = C_{rs}^t(y) (U_\gamma^{(r)} \otimes U_\gamma^{(s)}).$$

Infinitesimally this becomes the covariant constancy condition

$$\nabla C_{rs}^t = 0.$$

The connections acting on different multiplicity spaces are therefore constrained by the same network of fusion intertwiners and cannot be chosen independently.

This observation suggests a reconstruction principle. Let C_{rel} denote the category of localized relational sectors and their physical intertwiners, and let

$$\omega_x : C_{\text{rel}} \longrightarrow \text{Hilb}_{\text{fd}}$$

assign to each sector its multiplicity space at the event x . The group of unitary tensor automorphisms

$$G = \text{Aut}^\otimes(\omega)$$

preserves the full fusion structure. Under the standard rigidity and locality conditions familiar from Tannaka–Krein and Doplicher–Roberts reconstruction, such data determine a compact internal symmetry group [?, ?]. In the MS interpretation, the important question is not the abstract reconstruction theorem itself, but whether the projective comparison sectors generated by the underlying dynamics satisfy the physical localization, transportability, fusion, and duality properties required to supply its input. If they do, the connections on individual matter sectors take the associated-bundle form

$$A^{(r)} = \rho_{r*}(A)$$

for a single internal comparison connection A and a family of representations ρ_r .

The construction developed here therefore stops short of identifying an induced multiplicity connection with a Yang–Mills field. A Berry or Wilczek–Zee connection can be a composite connection determined entirely by slow relational order parameters. An independently propagating gauge field requires an additional infrared dynamical step: the comparison connection must acquire its own transverse low-energy mode and a positive kinetic stiffness. At quadratic order one may diagnose such a regime by an inverse propagator of the schematic form

$$\Gamma_{\mu\nu}^{(2)}(k) = \left[m_A^2 + K_A k^2 + O(k^4) \right] P_{\mu\nu}^T.$$

A massless dynamical gauge mode requires

$$m_A^2 = 0, \quad 0 < K_A < \infty.$$

Only in such a regime is it meaningful to identify K_A with an inverse gauge stiffness and to interpret the internal comparison connection as a propagating Yang–Mills degree of freedom.

The logical structure of the present paper is therefore deliberately narrower than a derivation of the Standard Model. We do not derive the groups $SU(3)$, $SU(2)$, or $U(1)$, nor chirality, spin-statistics, anomaly cancellation, or numerical coupling constants. We instead isolate a more primitive question: how can internal quantum degrees of freedom and their comparison structure arise when the starting point contains only geometric comparison and projective quantum carriers? The answer proposed here is that many-body projective composition produces geometry-invisible multiplicity spaces; non-commuting relational observables turn these spaces into genuine quantum systems; protected event-dependent multiplicities form internal bundles; and compatibility between their transport and fusion provides the structural input from which a common internal gauge group may be reconstructed. In this sense, internal quantum structure is not introduced as an additional set of labels attached to spacetime objects. It is identified with coherent information contained in the different ways in which the same geometric representation can be relationally realized.

2 Projective Carriers and Relational Multiplicity

We take as input a localized infrared sector carrying the nontrivial projective rotational class and assume that the lowest relevant irreducible carrier is the spin- $\frac{1}{2}$ representation

$$V_{\frac{1}{2}} \simeq \mathbb{C}^2.$$

This assumption is stronger than the statement $\Xi = -I$. The latter selects the half-integer projective class but allows all irreducible representations with

$$j \in \left\{ \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots \right\}.$$

A dynamical infrared minimality condition, an exact AKLT-type edge realization, or an equivalent mechanism is therefore required to isolate $j = \frac{1}{2}$. The analysis below concerns the consequences of such a minimal projective sector once it has been obtained. It does not derive that sector from the original local MS response axioms.

Consider N identical minimal projective carriers. Their Hilbert space is

$$\mathcal{H}_N = \left(V_{\frac{1}{2}}\right)^{\otimes N},$$

with collective rotational action

$$U_N(g) = U_{\frac{1}{2}}(g)^{\otimes N}, \quad g \in SU(2).$$

The standard Clebsch–Gordan decomposition gives

$$\mathcal{H}_N \simeq \bigoplus_j V_j \otimes M_j^{(N)},$$

where V_j is the irreducible spin- j representation and $M_j^{(N)}$ is its multiplicity space. The collective geometric action takes the block form

$$U_N(g) = \bigoplus_j U_j(g) \otimes I_{M_j^{(N)}}.$$

Thus the collective rotation acts on V_j but acts trivially on the multiplicity factor.

For spin- $\frac{1}{2}$ tensor powers, the multiplicity of V_j is

$$m_j^{(N)} := \dim M_j^{(N)} = \binom{N}{N/2-j} - \binom{N}{N/2-j-1},$$

where a binomial coefficient is understood to vanish when its lower argument is outside the interval $[0, N]$. The admissible values of j are the usual ones compatible with the parity of N :

$$j = \begin{cases} 0, 1, \dots, N/2, & N \text{ even}, \\ \frac{1}{2}, \frac{3}{2}, \dots, N/2, & N \text{ odd}. \end{cases}$$

For completeness, Eq. (2) can be obtained by counting states of fixed magnetic quantum number. Let $d_M^{(N)}$ denote the dimension of the subspace with total $J_z = M$. For $M \geq 0$,

$$d_M^{(N)} = \binom{N}{N/2-M}.$$

Every spin- j irreducible representation contributes one state to each magnetic subspace with $|M| \leq j$. Hence

$$d_j^{(N)} = \sum_{k \geq j} m_k^{(N)},$$

and subtraction of the neighboring equations gives

$$m_j^{(N)} = d_j^{(N)} - d_{j+1}^{(N)},$$

which yields Eq. (2).

The representation-theoretic content of Eq. (2) is standard. Its role in the present construction is operational. The collective $SU(2)$ action is the part of the projective carrier that transforms under rotational comparison. Once the geometric representation V_j is fixed, states that differ only in the factor $M_j^{(N)}$ cannot be distinguished by the collective rotational action itself. We therefore make the following definition.

Definition. For a fixed collective spin sector j , the space $M_j^{(N)}$ will be called the *relational multiplicity space*. It contains the coherent degrees of freedom that distinguish different realizations of the same collective geometric representation.

This terminology does not assert that every multiplicity degree of freedom is a fundamental internal charge. It identifies the part of the many-carrier Hilbert space that is invisible to the specified collective geometric action. Whether such degrees of freedom survive localization, acquire a stable low-energy dynamics, or become carriers of an internal gauge interaction is a separate question addressed later.

The first nontrivial example occurs already for three carriers:

$$\left(V_{\frac{1}{2}}\right)^{\otimes 3} \simeq V_{\frac{3}{2}} \oplus \left(V_{\frac{1}{2}} \otimes \mathbb{C}^2\right).$$

Thus the total spin- $\frac{1}{2}$ sector contains two independent factors:

$$V_{\frac{1}{2}}^{\text{geom}} \otimes M_{\frac{1}{2}}^{(3)}, \quad M_{\frac{1}{2}}^{(3)} \simeq \mathbb{C}^2.$$

The first factor carries the ordinary spin- $\frac{1}{2}$ action. The second factor is a two-dimensional state space on which the collective rotation acts as the identity. In this sense three minimal projective carriers already contain a geometric spin doublet together with an independent relational qubit.

Four carriers provide an even sharper example:

$$\left(V_{\frac{1}{2}}\right)^{\otimes 4} \simeq V_2 \oplus \left(V_1 \otimes \mathbb{C}^3\right) \oplus \left(V_0 \otimes \mathbb{C}^2\right).$$

The total-spin-zero sector is therefore

$$\mathcal{I}_4 := \text{Inv}_{SU(2)} \left[\left(V_{\frac{1}{2}}\right)^{\otimes 4} \right] \simeq \mathbb{C}^2.$$

Unlike the three-carrier example, the complete geometric representation in this sector is trivial:

$$U_4(g)|_{\mathcal{I}_4} = I_{\mathcal{I}_4} \quad \forall g \in SU(2).$$

The two-dimensional state space in Eq. (2) is therefore entirely invisible to collective rotational comparison.

A convenient orthonormal basis of $\mathcal{I}_4$ is obtained by first coupling the pairs (1, 2) and (3, 4). Let

$$|s\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

denote the two-spin singlet and let $|t_+\rangle$, $|t_0\rangle$, and $|t_-\rangle$ denote the spin-one triplet basis. Define

$$|0_L\rangle = |s_{12}\rangle|s_{34}\rangle$$

and

$$|1_L\rangle = \frac{1}{\sqrt{3}} (|t_{+,12}\rangle|t_{-,34}\rangle - |t_{0,12}\rangle|t_{0,34}\rangle + |t_{-,12}\rangle|t_{+,34}\rangle).$$

Both states have total spin zero and satisfy

$$U_{\frac{1}{2}}(g)^{\otimes 4}|a_L\rangle = |a_L\rangle, \quad a = 0, 1.$$

Consequently every normalized state

$$|\psi_L\rangle = \alpha|0_L\rangle + \beta|1_L\rangle, \quad |\alpha|^2 + |\beta|^2 = 1,$$

has identical collective rotational quantum numbers.

The existence of this multiplicity should not be confused with an accidental energy degeneracy. Equations (2)–(2) are statements about the representation of the geometric comparison group. They remain meaningful before a Hamiltonian is specified. A Hamiltonian may subsequently split, preserve, or dynamically mix the multiplicity states, but the collective geometric action cannot distinguish them.

The same distinction is encoded algebraically by the commutant. Let

$$\mathcal{A}_{\text{geom}}^{(N)}$$

denote the operator algebra generated by the collective $SU(2)$ action on $\mathcal{H}_N$. Its commutant is

$$\left(\mathcal{A}_{\text{geom}}^{(N)}\right)' \simeq \bigoplus_j I_{V_j} \otimes \text{End}\left(M_j^{(N)}\right).$$

Thus, within a fixed j sector, every operator acting only on the multiplicity factor commutes with collective geometry. The maximal unitary group available kinematically on that factor is

$$U\left(m_j^{(N)}\right).$$

At this stage this group is only a maximal basis group of the multiplicity space. It is not yet a physical gauge group. In particular, neither Eq. (2) nor the dimension of $M_j^{(N)}$ determines which transformations are dynamically realizable by comparison transport.

The decomposition also suggests a precise separation between two kinds of quantum information. Information encoded in V_j changes under collective rotational comparison and therefore carries geometric spin information. Information encoded in $M_j^{(N)}$ can vary while all collective rotational quantum numbers remain unchanged. We shall call the latter *geometry-invisible relational information*. The next section shows that this information supports noncommuting observables and is therefore not merely a classical degeneracy label.

3 Noncommutative Relational Observables

The existence of a multiplicity space does not by itself imply a genuinely quantum internal structure. A classical degeneracy could also produce several states with identical external quantum numbers. The decisive question is whether physically meaningful observables acting within the multiplicity sector form a noncommutative algebra.

For two minimal projective carriers define the projector onto their spin-singlet channel by

$$P_{ij}^{(0)} = \frac{1}{4} (I - \sigma_i \cdot \sigma_j),$$

where

$$\sigma_i \cdot \sigma_j = \sum_{a=1}^3 \sigma_i^a \sigma_j^a.$$

Equivalently, in terms of $\mathbf{J}_i = \frac{1}{2}\boldsymbol{\sigma}_i$ with $\hbar = 1$ in this representation-theoretic normalization,

$$P_{ij}^{(0)} = \frac{1}{4}I - \mathbf{J}_i \cdot \mathbf{J}_j.$$

The operator $P_{ij}^{(0)}$ is a rotational scalar:

$$[P_{ij}^{(0)}, \mathbf{J}_{\text{tot}}] = 0, \quad \mathbf{J}_{\text{tot}} = \sum_{k=1}^N \mathbf{J}_k.$$

It therefore belongs to the commutant of the collective geometric action.

Operationally, $P_{ij}^{(0)}$ asks a relational question. It does not measure a direction carried by either constituent separately. It distinguishes whether the pair (i, j) occupies its neutral rotational channel. Different pairings correspond to different decompositions of the same multi-carrier state, and these relational questions need not be compatible.

This can be seen explicitly in the four-carrier invariant space $\mathcal{I}_4$. In the basis $\{|0_L\rangle, |1_L\rangle\}$ defined by Eqs. (2) and (2), the projector onto the $(1, 2)$ singlet channel is

$$P_{12}^{(0)}|_{\mathcal{I}_4} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Recoupling the four spins gives

$$P_{23}^{(0)}|_{\mathcal{I}_4} = \begin{pmatrix} \frac{1}{4} & -\frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}.$$

Their commutator is

$$[P_{12}^{(0)}, P_{23}^{(0)}] = \frac{\sqrt{3}}{4} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \neq 0.$$

Hence the two propositions

the pair $(1, 2)$ is in its neutral channel

and

the pair $(2, 3)$ is in its neutral channel

cannot be represented as simultaneously sharp classical properties of the relational qubit.

The result can be stated independently of the chosen logical basis.

Proposition 1 (Relational incompatibility). Let $\mathcal{I}_4$ be the total-spin-zero sector of four spin- $\frac{1}{2}$ projective carriers. The restrictions of the pair-neutrality projectors $P_{12}^{(0)}$ and $P_{23}^{(0)}$ to $\mathcal{I}_4$ do not commute. Therefore the algebra of rotationally invariant pair relations on $\mathcal{I}_4$ is noncommutative.

Proof. Equations (3) and (3) give the explicit restrictions in an orthonormal basis of $\mathcal{I}_4$, and Eq. (3) is nonzero. $\square$

This noncommutativity is entirely internal to the relational multiplicity space. Both operators commute with the collective $SU(2)$ action, and every state in $\mathcal{I}_4$ has the same total geometric representation $j = 0$. The incompatibility is therefore not inherited from choosing two different spatial axes for an ordinary spin measurement. It concerns two different ways of organizing neutral pair relations inside a state that is itself a collective rotational scalar.

More strongly, the pair-neutrality operators generate the complete operator algebra of the relational qubit. Define logical Pauli operators by

$$Z_L = 2P_{12}^{(0)} - I.$$

Equation (3) gives

$$P_{23}^{(0)} = \frac{1}{2}I - \frac{1}{4}Z_L - \frac{\sqrt{3}}{4}X_L,$$

and therefore

$$X_L = -\frac{4}{\sqrt{3}}\left(P_{23}^{(0)} - \frac{1}{2}I + \frac{1}{4}Z_L\right).$$

The remaining logical Pauli operator follows from the commutator,

$$Y_L = \frac{1}{2i}[Z_L, X_L].$$

Thus the unital $*$ -algebra generated by $P_{12}^{(0)}$ and $P_{23}^{(0)}$ contains I , X_L , Y_L , and Z_L .

Theorem 2 (Complete relational qubit algebra). On the four-carrier invariant space $\mathcal{I}_4$,

$$\text{Alg}^*(P_{12}^{(0)}, P_{23}^{(0)}) = \text{End}(\mathcal{I}_4) \simeq M_2(\mathbb{C}).$$

Consequently, complete qubit dynamics can in principle be generated using operators that are rotational scalars at the level of the constituent carriers.

Proof. Equations (3)–(3) show that the algebra generated by the two projectors contains a basis of $M_2(\mathbb{C})$. $\square$

Theorem 2 provides a precise meaning for a quantum degree of freedom that is internal to relational structure. A general logical Hamiltonian

$$H_L = aI + bX_L + cY_L + dZ_L$$

can be constructed from rotationally invariant relational operators and their compositions. Such a Hamiltonian may generate nontrivial unitary evolution in $\mathcal{I}_4$ while every state remains a total rotational singlet.

The same phenomenon occurs in the three-carrier spin- $\frac{1}{2}$ sector. There the decomposition

$$V_{\frac{1}{2}} \otimes M_{\frac{1}{2}}^{(3)}$$

contains a two-dimensional multiplicity space. Pair-singlet projectors act trivially on neither copy of $M_{\frac{1}{2}}^{(3)}$ and furnish noncommuting operators on this factor. The distinction from the four-carrier case is

that the full physical sector still carries the nontrivial geometric factor $V_{\frac{1}{2}}$. The three-carrier system therefore provides a minimal example of the tensor-product structure

$$\mathcal{H}_{\text{sector}} \simeq V_{\text{geom}} \otimes M_{\text{rel}},$$

whereas the four-carrier invariant sector isolates the relational factor by setting

$$V_{\text{geom}} = V_0 \simeq \mathbb{C}.$$

There is also a useful recoupling interpretation of Eq. (3). The state $|s_{12}\rangle|s_{34}\rangle$ and the state obtained by forming singlets in a different pairing, such as $|s_{14}\rangle|s_{23}\rangle$, are not orthogonal classical configurations. For example, their overlap has magnitude

$$|\langle s_{12}s_{34}|s_{14}s_{23}\rangle| = \frac{1}{2}.$$

Accordingly, different pairing patterns cannot be treated as mutually exclusive hidden graphs carried by the same four constituents. They are different, nonorthogonal states in the same two-dimensional invariant Hilbert space.

This observation rules out a purely classical interpretation of the relational multiplicity. A relation-first description would remain classical if every pair relation possessed a simultaneous context-independent value and quantum uncertainty represented only ignorance of that underlying network. Equations (3) and (3) instead imply that different relational decompositions themselves constitute incompatible observables. The quantum structure therefore resides not merely in uncertainty about which relation is present, but in the noncommutative algebra of possible relational questions.

The algebra generated by the pair projectors is closely related to familiar centralizer structures in spin systems. If one introduces

$$e_i = 2P_{i,i+1}^{(0)},$$

then for spin- $\frac{1}{2}$ carriers these operators satisfy the Temperley–Lieb relations at loop parameter $\delta = 2$,

$$e_i^2 = 2e_i, \quad e_i e_{i\pm 1} e_i = e_i, \quad [e_i, e_j] = 0 \quad \text{for } |i - j| > 1.$$

Equivalently, the swap operator

$$\Pi_{ij} = I - 2P_{ij}^{(0)} = \frac{1}{2}(I + \sigma_i \cdot \sigma_j)$$

belongs to the same rotational commutant. The symmetric-group action generated by such exchanges provides the standard centralizer counterpart to the collective $SU(2)$ action. These representation-theoretic facts are not new. Their significance here is that the centralizer generators have a direct comparison interpretation: they alter the way projective carriers are relationally coupled while preserving their collective geometric representation.

It is useful to separate three statements that are sometimes conflated. The first is representation-theoretic:

$$\mathcal{H}_N = \bigoplus_j V_j \otimes M_j^{(N)}.$$

The second is algebraic:

$$\text{End}_{SU(2)}(\mathcal{H}_N) = \bigoplus_j I_{V_j} \otimes \text{End}(M_j^{(N)}).$$

The third is physical and is the claim advanced in the present framework: when the $SU(2)$ action is identified with the collective geometric comparison action, the noncommuting operators on $M_j^{(N)}$ represent coherent transformations among different relational realizations of the same geometric state. Only the third statement is specific to the MS interpretation.

This distinction will be essential in the following sections. The existence of the full matrix algebra $\text{End}(M_j^{(N)})$ determines the kinematic space of possible internal relational observables, but it does not determine which of them are generated by the low-energy Hamiltonian, which remain exactly degenerate, or which can be transported consistently between spacetime events. In particular,

$$\text{End}(M_j^{(N)})$$

is not yet a gauge algebra, and

$$U(m_j^{(N)})$$

is not yet a physical gauge group. The transition from a noncommutative relational multiplicity to an internal comparison bundle requires additional dynamical and locality structure.

The next step is therefore to ask how much information can be stored in these geometry-invisible sectors and how strongly they are protected against one-body geometric probes. This will provide the physical criterion needed before promoting the multiplicity spaces to event-dependent internal bundles.

4 Geometric Blindness and the Capacity of Neutral Relational States

The previous section established that rotationally invariant pair relations generate a genuinely noncommutative algebra on the multiplicity space. We now ask in what precise sense the information stored in such a space is internal rather than merely degenerate under collective rotation. Two properties are particularly relevant. First, a total-spin-neutral relational state is completely invisible to one-carrier rotational tomography. Second, local geometric vector perturbations possess no matrix elements within the neutral sector at first order. Together these properties distinguish relational multiplicity from an ordinary collection of externally visible spin degrees of freedom.

Let

$$\mathcal{I}_N := \text{Inv}_{SU(2)} \left[\left(V_{\frac{1}{2}} \right)^{\otimes N} \right]$$

denote the total-spin-zero sector for even N . Let $\rho \in \text{End}(\mathcal{I}_N)$ be any density operator supported entirely in this sector. Since the collective action is trivial on $\mathcal{I}_N$,

$$U_{\frac{1}{2}}(g)^{\otimes N} \rho U_{\frac{1}{2}}(g)^{\otimes N\dagger} = \rho \quad \forall g \in SU(2).$$

Taking the partial trace over all carriers except the i th gives

$$U_{\frac{1}{2}}(g) \rho_i U_{\frac{1}{2}}(g)^\dagger = \rho_i, \quad \rho_i = \text{Tr}_{\hat{i}} \rho.$$

Because $V_{\frac{1}{2}}$ is irreducible, Schur's lemma implies that ρ_i is proportional to the identity. Unit trace fixes the coefficient.

Theorem 3 (One-carrier geometric blindness). For every density operator supported in the rotationally invariant sector $\mathcal{I}_N$,

$$\rho_i = \frac{1}{2}I$$

for every constituent carrier i .

Proof. Equation (4) shows that ρ_i commutes with the irreducible spin- $\frac{1}{2}$ representation of $SU(2)$. Hence $\rho_i = cI$ by Schur's lemma. The normalization $\text{Tr} \rho_i = 1$ gives $c = \frac{1}{2}$. $\square$

The theorem means that no measurement performed on one projective carrier alone can distinguish two states stored purely in $\mathcal{I}_N$. In particular, for the four-carrier logical state

$$|\psi_L\rangle = \alpha|0_L\rangle + \beta|1_L\rangle,$$

every individual reduced state is independent of α and β :

$$\rho_1 = \rho_2 = \rho_3 = \rho_4 = \frac{1}{2}I.$$

The logical amplitudes therefore reside in correlations among the carriers rather than in any one-carrier rotational degree of freedom.

This property is related to familiar noiseless-subsystem and decoherence-free-subspace constructions for collective symmetry [?, ?]. The present use is more specific. The collective symmetry is identified with the rotational comparison action inherited from the MS geometric sector, and the invariant or multiplicity degrees of freedom are interpreted as coherent relational information left unresolved by that geometric action.

One-carrier blindness at the level of reduced states has a corresponding operator statement. Let

$$P_{\text{inv}}$$

denote the orthogonal projector onto $\mathcal{I}_N$. The local spin operator $\mathbf{J}_i$ transforms as a vector under collective rotation:

$$U_N(g)J_i^a U_N(g)^{-1} = R(g)^a_b J_i^b,$$

where $R(g) \in SO(3)$ is the corresponding vector rotation. Define

$$A_i^a := P_{\text{inv}} J_i^a P_{\text{inv}}.$$

Since the collective representation acts trivially on the invariant sector, Eq. (4) gives

$$A_i^a = R(g)^a_b A_i^b \quad \forall g \in SU(2).$$

There is no nonzero invariant vector in the ordinary three-dimensional rotation representation. Hence all three operators A_i^a vanish.

Proposition 4 (First-order one-body geometric blindness). For every carrier i and spatial component a ,

$$P_{\text{inv}} J_i^a P_{\text{inv}} = 0.$$

Consequently, every one-body vector perturbation

$$V = \sum_i \Omega_i^a J_i^a$$

satisfies

$$P_{\text{inv}} V P_{\text{inv}} = 0.$$

This is stronger than the statement that the expectation value of a single spin vanishes. Equation (4) says that no one-body vector operator has any matrix element between two states of the neutral relational sector. A local vector perturbation therefore cannot directly implement a logical operation within $\mathcal{I}_N$ at first order. It can instead couple the neutral sector to nonzero-total-spin sectors.

Suppose that the neutral subspace is separated from such sectors by a positive gap and let

$$P = P_{\text{inv}}, \quad Q = I - P.$$

If the unperturbed energy of the neutral sector is E_0 , standard degenerate perturbation theory gives the leading nontrivial effective operator

$$H_{\text{eff}}^{(2)} = -P V Q (H_Q - E_0)^{-1} Q V P.$$

Thus the leading internal response to a local geometric vector perturbation is second order.

A particularly transparent form is obtained when the relevant excited sector may be approximated by a common gap $\Delta > 0$. Then

$$H_{\text{eff}}^{(2)} \simeq -\frac{1}{\Delta} P V^2 P.$$

Rotational invariance restricts the projected product of two vector operators. For singlet-to-singlet matrix elements,

$$P J_i^a J_j^b P = \frac{\delta^{ab}}{3} P (\mathbf{J}_i \cdot \mathbf{J}_j) P.$$

Substitution into Eq. (4) yields

$$H_{\text{eff}}^{(2)} \simeq -\frac{1}{3\Delta} \sum_{i,j} (\boldsymbol{\Omega}_i \cdot \boldsymbol{\Omega}_j) P (\mathbf{J}_i \cdot \mathbf{J}_j) P.$$

The induced operator is therefore built from rotational scalars and acts directly on relational multiplicity.

Equation (4) gives a useful structural interpretation. A one-body geometric disturbance is incapable of resolving neutral relational information linearly, but virtual excursions outside the neutral sector generate effective pair relations. Geometry couples to the internal sector through differential or composite comparison rather than through a direct one-body label.

This motivates the following terminology.

Relational geometric blindness principle. A neutral multiplicity sector is geometrically blind, relative to the collective rotational comparison algebra, when collective transformations act trivially on the sector and rank-one local geometric perturbations have no first-order matrix elements within it. Its leading response to such perturbations is then mediated by relational scalar operators or by leakage through nonneutral sectors.

The qualifier “relative to the collective rotational comparison algebra” is essential. The statement does not imply that the relational sector is invisible to every possible gravitational, spacetime, or microscopic probe. Indeed, the pair observables developed in Section 3 distinguish states within the sector precisely because they are relational rather than one-body rotational observables.

The neutral relational state space also grows rapidly with the number of carriers. For

$$N = 2n,$$

the multiplicity of the spin-zero representation follows from Eq. (2):

$$\dim \mathcal{I}_{2n} = \binom{2n}{n} - \binom{2n}{n-1}.$$

Using

$$\binom{2n}{n-1} = \frac{n}{n+1} \binom{2n}{n},$$

one obtains

$$\dim \mathcal{I}_{2n} = \frac{1}{n+1} \binom{2n}{n} = C_n,$$

where C_n is the n th Catalan number.

Its large- n asymptotics is

$$C_n \sim \frac{4^n}{\sqrt{\pi} n^{3/2}}.$$

Since $N = 2n$,

$$\log_2 \dim \mathcal{I}_N = N - \frac{3}{2} \log_2 N + O(1).$$

Theorem 5 (Asymptotically extensive neutral relational capacity). Let

$$k_N := \log_2 \dim \mathcal{I}_N$$

denote the logarithmic quantum-information capacity of the total-spin-zero sector of N spin- $\frac{1}{2}$ carriers, with N even. Then

$$k_N = N - \frac{3}{2} \log_2 N + O(1)$$

and therefore

$$\lim_{N \rightarrow \infty} \frac{k_N}{N} = 1.$$

The theorem shows that imposing complete collective rotational neutrality removes only a subextensive amount of Hilbert-space capacity. A large relational network can therefore carry an exponentially large internal state space while possessing total geometric spin zero. The phrase “internal state space” is used here in the representation-theoretic sense developed above: the information is invisible to collective rotation and inaccessible to one-carrier tomography, but it remains available to noncommuting multi-carrier relational observables.

This result also prevents an overly restrictive interpretation of geometric neutrality. The condition

$$\mathbf{J}_{\text{tot}} = 0$$

does not imply an absence of quantum structure. On the contrary, $\mathcal{I}_N$ supports an asymptotically extensive family of coherent states and operators. The external geometric representation can therefore remain simple while the relational multiplicity becomes increasingly complex.

The combination of Eqs. (4), (4), and (4) provides the physical motivation for treating multiplicity spaces as candidates for internal quantum carriers. They are not only invariant under the collective geometric action; they can store a large amount of coherent information that is absent from every individual constituent and is protected against direct one-body vector action at first order. The remaining question is how such spaces can be compared between different spacetime events. We now turn from a multiplicity space at one event to a multiplicity bundle over spacetime.

5 From Relational Multiplicity to Internal Bundles

A multiplicity space at a single event is not yet an internal gauge structure. Gauge comparison requires a family of internal state spaces over spacetime together with a rule for comparing them at different events. The first step is therefore to promote a protected relational multiplicity to a smooth event-dependent subspace of a larger comparison Hilbert bundle.

Let

$$\mathcal{R} \longrightarrow M$$

be a finite-rank Hermitian vector bundle representing an ambient local comparison space, and let

$$D : \Gamma(\mathcal{R}) \longrightarrow \Omega^1(M, \mathcal{R})$$

be a unitary ambient comparison connection. We do not assume that $\mathcal{R}$ is itself an internal gauge bundle. It is only a carrier large enough to contain the relevant low-energy relational states.

Suppose that the local relational Hamiltonian, constraint, or response kernel selects a smooth rank- m spectral projector

$$P(x) : \mathcal{R}_x \longrightarrow \mathcal{R}_x, \quad P(x)^2 = P(x), \quad P(x)^\dagger = P(x),$$

whose image is separated from the remaining local spectrum by a nonzero gap. The protected relational subspace is

$$\mathcal{M}_x := \text{im } P(x).$$

If the rank remains constant, these fibers define a Hermitian vector bundle

$$\mathcal{M} \longrightarrow M.$$

Locally choose an orthonormal frame

$$Z(x) = (|m_1(x)\rangle, \dots, |m_m(x)\rangle),$$

viewed as an isometric map

$$Z(x) : \mathbb{C}^m \longrightarrow \mathcal{R}_x.$$

It satisfies

$$Z^\dagger Z = I_m, \quad ZZ^\dagger = P.$$

A different local orthonormal basis of the same physical subspace has the form

$$Z(x) \longmapsto Z(x)U(x), \quad U(x) \in U(m).$$

This transformation changes the coordinates used inside $\mathcal{M}_x$ but does not alter the physical projector:

$$P \mapsto P.$$

The ambient derivative induces a natural connection on the protected subbundle,

$$\nabla^{\mathcal{M}} = PD$$

when acting on sections of $\mathcal{M}$. In the frame Z , its matrix-valued connection one-form is

$$A = Z^\dagger DZ.$$

Under Eq. (5),

$$A \mapsto U^{-1}AU + U^{-1}dU.$$

The familiar form of a $U(m)$ gauge transformation thus appears as the change of local relational frame inside a protected multiplicity bundle.

It is important, however, not to interpret Eq. (5) too quickly. A changing basis alone does not produce physical internal curvature.

Proposition 6 (Fixed-subspace no-go). Assume that the ambient comparison bundle is locally trivial with flat connection $D = d$, and suppose that the physical relational projector is constant,

$$P(x) = P_0.$$

Then every local relational frame has the form

$$Z(x) = Z_0 U(x), \quad U(x) \in U(m),$$

and the induced connection is

$$A = U^{-1}dU.$$

Consequently,

$$F_A = dA + A \wedge A = 0.$$

Proof. A constant projector defines a fixed m -plane. Every orthonormal frame of that plane differs from a fixed frame Z_0 by a unitary matrix $U(x)$. Substituting $Z = Z_0 U$ into Eq. (5) gives $A = U^{-1}dU$, whose curvature vanishes by the Maurer–Cartan identity. $\square$

Thus

$$\dim \mathcal{M}_x = m$$

and the associated freedom to choose a $U(m)$ basis are insufficient to produce an internal field strength. Nonzero curvature generated by the relational sector requires the physical low-energy subspace itself to vary inside the ambient comparison space. In a nonflat ambient bundle the corresponding statement is that a D -parallel projector contributes no new extrinsic relational curvature; the induced curvature is then only the projection of the ambient curvature.

The geometry of the moving subspace is encoded by the covariant derivative of the projector. Let

$$Q = I - P$$

and define the extrinsic relational derivative

$$K := QDZ.$$

The derivative of the local frame decomposes into tangential and normal components,

$$DZ = ZA + K.$$

For a physical state written locally as

$$|\Psi\rangle = Z\psi, \quad \psi \in \mathbb{C}^m,$$

one obtains

$$D|\Psi\rangle = Z(d + A)\psi + K\psi.$$

The first term remains inside the protected relational sector and defines its internal comparison. The second term points into the orthogonal complement and measures leakage or extrinsic motion of the low-energy subspace.

This decomposition provides a precise distinction between internal transport and changes of the physical embedding. The matrix A compares coordinates within the protected multiplicity bundle, whereas K measures how the protected bundle itself bends inside the larger comparison space.

The curvature of the induced connection follows directly from the projected connection construction.

Theorem 7 (Relational Gauss curvature identity). Let $F_D = D^2$ denote the curvature of the ambient comparison connection. The curvature of the induced connection on $\mathcal{M}$ is

$$F_{\mathcal{M}} = PF_DP + P(DP \wedge DP)P.$$

In a local orthonormal frame Z this becomes

$$F_A = Z^\dagger F_D Z + K^\dagger \wedge K,$$

up to the equivalent sign convention obtained by choosing Hermitian rather than anti-Hermitian gauge potentials.

In particular, if the ambient comparison connection is flat,

$$F_D = 0,$$

then

$$F_A = K^\dagger \wedge K.$$

The theorem has a direct interpretation within the relational construction. Internal curvature need not be identified with spacetime curvature and need not already exist as an independent microscopic gauge field. It may arise because the protected low-energy relational subspace twists nontrivially inside a larger comparison Hilbert space.

Writing

$$K = K_\mu dx^\mu,$$

one may define the matrix-valued quantum-geometric tensor

$$Q_{\mu\nu} := K_\mu^\dagger K_\nu.$$

For a flat ambient connection,

$$F_{\mu\nu} = Q_{\mu\nu} - Q_{\nu\mu}.$$

The Hermitian symmetric part of $Q_{\mu\nu}$ measures the local distinguishability of neighboring protected subspaces, whereas its antisymmetric part determines their induced internal curvature. Both structures therefore arise from the same extrinsic response of the relational projector.

A simple norm estimate follows immediately:

$$\|F_{\mu\nu}\|_{\text{HS}} \leq 2\|K_\mu\|_{\text{HS}}\|K_\nu\|_{\text{HS}}.$$

Defining

$$g_{\mu\mu}^{\text{rel}} = \text{Tr}(K_\mu^\dagger K_\mu),$$

one obtains

$$\|F_{\mu\nu}\|_{\text{HS}} \leq 2\sqrt{g_{\mu\mu}^{\text{rel}}g_{\nu\nu}^{\text{rel}}}.$$

Thus a large induced internal curvature requires a nonzero susceptibility of the protected relational subspace to the corresponding comparison deformations. A relational sector that is completely rigid under all such deformations cannot simultaneously support nontrivial extrinsic holonomy.

When the ambient comparison fiber is locally identified with $\mathbb{C}^N$, a rank- m projector specifies a point of the complex Grassmannian,

$$P(x) \in \text{Gr}(m, N).$$

The bundle $\mathcal{M}$ is then the pullback of the tautological rank- m bundle over $\text{Gr}(m, N)$. Event-dependent relational vacua can therefore be represented by maps

$$P : M \longrightarrow \text{Gr}(m, N).$$

This formulation separates the physical object P from the local frame Z : the Grassmannian point specifies the actual protected subspace, whereas the $U(m)$ transformation of Z changes only its local coordinates.

For fixed m and increasing ambient dimension, the stable Grassmannian approaches the classifying space

$$\text{Gr}(m, \infty) \simeq BU(m).$$

Consequently, sufficiently large complex relational comparison spaces possess the topological capacity required to realize general rank- m complex vector bundles. This statement is purely kinematic. It does not imply that the MS dynamics realizes every element of $BU(m)$, nor does it select a physical gauge group. It shows only that an exponentially large relational multiplicity space, such as those found in Section 4, is not topologically too small to serve as an ambient carrier for nontrivial internal bundles.

The existence of the vector bundle $\mathcal{M}$ must also be distinguished from exact internal degeneracy. Suppose the local effective Hamiltonian restricted to the protected subspace is

$$PH_{\text{rel}}P = E(x)I_m + \Phi(x),$$

with Φ traceless. If

$$\Phi = 0,$$

the m states remain locally degenerate and the full $U(m)$ change of frame can be treated as a redundancy of the low-energy description. If

$$\Phi \neq 0,$$

the eigenstructure of Φ selects preferred directions inside the multiplicity space. The symmetry preserved by the dynamics is then the stabilizer of Φ , generally a proper subgroup of $U(m)$. A multiplicity splitting is therefore better interpreted as an internal order parameter than as a gauge connection.

This distinction can be summarized as follows:

$$\text{motion of the protected subspace} \longrightarrow \text{connection and curvature,}$$

whereas

$$\text{splitting within the protected subspace} \longrightarrow \text{internal symmetry reduction.}$$

Both may coexist, but they describe different physical structures.

The construction of this section establishes an event-dependent internal comparison geometry, but it still does not determine the physical comparison group. The maximal local frame freedom $U(m)$ is only a kinematic envelope. The group actually generated by transport is determined by the curvature and its holonomy, and may be Abelian even when $m > 1$. The next section therefore turns from the existence of multiplicity bundles to the dynamical selection of their holonomy group.

6 Curvature, Holonomy, and the Selection of the Internal Comparison Group

The construction of Section 5 associates a unitary connection with an event-dependent relational multiplicity bundle. The existence of a rank- m bundle, however, does not determine the physical internal comparison group. The full unitary group $U(m)$ describes the maximal freedom to choose an orthonormal frame in an m -dimensional complex fiber. It need not be generated by the actual comparison dynamics.

The relevant dynamical object is instead the holonomy of the induced connection. Fix a base event $x \in M$ and let

$$\text{Hol}_x(\nabla^M) \subseteq U(\mathcal{M}_x)$$

denote the group generated by parallel transport around closed loops based at x . We define the dynamically realized internal comparison group by

$$G_{\text{int}}^{\text{dyn}}(x) := \overline{\text{Hol}_x(\nabla^M)}.$$

The closure is included so that the comparison transformations form a closed unitary subgroup. For base points in the same connected component, different choices of x give conjugate subgroups.

Equation (6) should be contrasted with the kinematic relation

$$G_{\text{kin}}(x) = U(m).$$

The latter is fixed by the fiber dimension. The former is determined by the connection. In general,

$$G_{\text{int}}^{\text{dyn}}(x) \subseteq G_{\text{kin}}(x),$$

and the inclusion may be strict.

The infinitesimal content of this distinction is controlled by curvature. Let

$$F_M \in \Omega^2(M, \text{End } \mathcal{M})$$

be the curvature of the multiplicity connection. For a path $\gamma : y \rightarrow x$, let

$$U_\gamma : \mathcal{M}_y \longrightarrow \mathcal{M}_x$$

denote parallel transport. The Ambrose–Singer theorem implies that the Lie algebra of the restricted holonomy group is generated by curvature values transported to a common fiber [?, ?]. In the present notation,

$$\mathfrak{g}_{\text{hol}}(x) = \text{Lie} \left\langle U_\gamma F_y(X, Y) U_\gamma^{-1} \right\rangle,$$

where y ranges over points connected to x , $X, Y \in T_y M$, and the brackets denote the smallest real Lie algebra containing all such transported curvature operators.

This provides the appropriate selection principle for the internal comparison group.

Proposition 8 (Curvature Lie-closure criterion). Let $\mathcal{M} \rightarrow M$ be a protected rank- m relational multiplicity bundle equipped with its induced unitary connection. The connected internal comparison group is determined by the Lie algebra generated by the parallel-transported curvature values in Eq. (6). In particular, the fiber dimension m fixes only the upper bound

$$\mathfrak{g}_{\text{hol}} \subseteq \mathfrak{u}(m)$$

and does not determine equality.

The proposition is mathematically a direct application of the standard holonomy theorem. Its role in the present reconstruction is conceptual: internal symmetry is not inferred from multiplicity counting. It must be selected by the curvature generated by the actual comparison dynamics.

The distinction becomes particularly transparent for a two-dimensional multiplicity space. Write the anti-Hermitian curvature as

$$F = \frac{i}{2} (f^0 I + f^a \sigma_a),$$

where f^0 and f^a are real two-forms. After parallel transport to a common fiber, the traceless curvature values may be identified with vectors in $\mathbb{R}^3$ through

$$\frac{i}{2} v^a \sigma_a \longleftrightarrow \mathbf{v}.$$

For two such matrices,

$$X = \frac{i}{2} \mathbf{u} \cdot \boldsymbol{\sigma}, \quad Y = \frac{i}{2} \mathbf{v} \cdot \boldsymbol{\sigma},$$

their commutator is

$$[X, Y] = -\frac{i}{2} (\mathbf{u} \times \mathbf{v}) \cdot \boldsymbol{\sigma}.$$

Hence two nonparallel traceless curvature directions generate a third independent direction through their commutator.

Corollary 9 (Rank-two non-Abelianity test). For a rank-two complex relational bundle, let $\mathcal{F}_x$ denote the set of all traceless curvature values transported to a common fiber $\mathcal{M}_x$.

If every nonzero element of $\mathcal{F}_x$ is proportional to a single traceless generator, then the connected noncentral holonomy is at most $U(1)$.

If $\mathcal{F}_x$ contains two noncommuting traceless elements, then

$$\mathfrak{su}(2) \subseteq \mathfrak{g}_{\text{hol}}.$$

Since the traceless algebra of $U(2)$ is already $\mathfrak{su}(2)$, the traceless connected holonomy is then the full $SU(2)$ algebra. If, in addition, the curvature contains an independent central component proportional to iI , then

$$\mathfrak{g}_{\text{hol}} = \mathfrak{u}(2).$$

Proof. If all transported traceless curvature operators are collinear, their commutators vanish and their Lie closure is one dimensional. If two traceless curvature operators fail to commute, their corresponding vectors $\mathbf{u}$ and $\mathbf{v}$ are nonparallel. Equation (6) then produces a third direction proportional to $\mathbf{u} \times \mathbf{v}$, and the three matrices span $\mathfrak{su}(2)$. A nonzero independent central generator commutes with this algebra and enlarges it to $\mathfrak{u}(2)$. $\square$

The corollary prevents an identification that would otherwise be tempting in the examples of Section 2: a relational qubit does not automatically imply an $SU(2)$ gauge interaction. A two-dimensional fiber provides the matrix algebra $M_2(\mathbb{C})$ and the maximal frame group $U(2)$, but the connection may explore only an Abelian subalgebra. Genuine non-Abelian comparison requires noncommuting curvature directions.

A minimal model makes the distinction explicit. Let the ambient relational space be

$$\mathcal{R} \simeq \mathbb{C}^3$$

and let a local comparison Hamiltonian select one normalized “bright” relational direction

$$|b\rangle \in \mathbb{C}^3, \quad \langle b|b\rangle = 1,$$

through

$$H_{\text{rel}} = E_D I_3 + \Delta |b\rangle\langle b|, \quad \Delta > 0.$$

The bright state has energy $E_D + \Delta$, whereas its orthogonal complement

$$\mathcal{D}_b = \{|\psi\rangle \in \mathbb{C}^3 : \langle b|\psi\rangle = 0\}$$

is a two-dimensional degenerate low-energy sector with energy E_D .

The corresponding projector is

$$P_D = I_3 - |b\rangle\langle b|.$$

Because the Hamiltonian depends only on the ray of $|b\rangle$, the parameter space of distinct bright directions is

$$\mathbb{CP}^2.$$

The dark doublets therefore form a rank-two complex bundle

$$\mathcal{D} \longrightarrow \mathbb{CP}^2.$$

This model realizes precisely the combination needed for a nontrivial multiplicity connection: the low-energy states remain locally degenerate, but their embedding inside the ambient comparison space changes with the physical bright direction. The degeneracy supplies a rank-two internal carrier, whereas the varying projector supplies curvature.

Choose the standard coordinate patch

$$|b(z)\rangle = \frac{(z_1, z_2, 1)^T}{\sqrt{1 + |z_1|^2 + |z_2|^2}}, \quad z_1, z_2 \in \mathbb{C}.$$

At

$$z_1 = z_2 = 0,$$

one has

$$|b\rangle = e_3, \quad \mathcal{D}_b = \text{span}\{e_1, e_2\}.$$

For a flat ambient comparison derivative, the projector curvature is

$$F_D = P_D (dP_D \wedge dP_D) P_D.$$

At the origin of the chosen patch its restriction to the dark doublet is

$$F_D = \begin{pmatrix} dz_1 \wedge d\bar{z}_1 & dz_1 \wedge d\bar{z}_2 \\ dz_2 \wedge d\bar{z}_1 & dz_2 \wedge d\bar{z}_2 \end{pmatrix},$$

up to the conventional overall factor associated with the choice of anti-Hermitian connection.

The curvature contains both diagonal and off-diagonal matrix directions. By evaluating Eq. (6) on independent real tangent two-planes, one obtains generators whose traceless parts are not mutually commuting. Their Lie closure therefore contains

$$\mathfrak{su}(2),$$

while the trace component provides an independent central direction. Hence the generic connected holonomy of this canonical dark-doublet bundle is

$$\mathfrak{g}_{\text{hol}} = \mathfrak{u}(2).$$

Proposition 10 (Minimal dark-doublet realization). The rank-two dark bundle defined by Eq. (6) over the full complex bright-state parameter space $\mathbb{CP}^2$ possesses noncommuting curvature directions. For the canonical projected connection its holonomy algebra is generically $\mathfrak{u}(2)$.

The result illustrates an important point. Non-Abelianity does not arise because the dark sector happens to have dimension two. It arises because the physical two-plane moves through sufficiently many complex directions in the ambient three-dimensional comparison space for the curvature to generate noncommuting endomorphisms.

The complex nature of the parameter space is also essential in this minimal example. Suppose instead that the microscopic relational Hamiltonian can be chosen real in a fixed ambient basis for all admissible comparison configurations. The bright direction is then restricted to a real ray,

$$[b] \in \mathbb{RP}^2,$$

and the dark space becomes a real rank-two bundle. Its orthonormal frame group is

$$O(2),$$

whose connected component is

$$SO(2) \simeq U(1).$$

Thus the same rank counting no longer supports a non-Abelian connected holonomy.

Corollary 11 (Complex-coherence requirement in the minimal bright-line model). For the rank-two dark sector obtained as the orthogonal complement of a single bright direction in a three-dimensional ambient space, restriction to a globally real comparison family reduces the connected frame group to $SO(2)$. Access to genuinely complex coherent comparison directions is required for the canonical model to realize the full non-Abelian $\mathfrak{u}(2)$ curvature algebra.

This corollary should not be interpreted as a general theorem that every non-Abelian emergent gauge structure requires the particular $\mathbb{CP}^2$ construction. Real vector bundles of higher rank can of course support non-Abelian orthogonal holonomy. The statement concerns the minimal rank-two dark-doublet realization considered here. It nevertheless shows that the complex quantum structure of the relational carrier can have direct consequences for the class of internal comparison groups available in low-energy dynamics.

The $\mathbb{CP}^2$ model also provides a useful warning against overextending a successful local holonomy construction. Let L denote the tautological bright line bundle over $\mathbb{CP}^2$. The trivial ambient bundle splits as

$$\underline{\mathbb{C}}^3 \simeq L \oplus \mathcal{D}.$$

If

$$h \in H^2(\mathbb{CP}^2, \mathbb{Z})$$

is the standard generator and the convention

$$c_1(L) = -h$$

is adopted, then

$$c(L) = 1 - h.$$

Since the ambient bundle is trivial,

$$c(L)c(\mathcal{D}) = 1,$$

and therefore

$$c(\mathcal{D}) = \frac{1}{1 - h} = 1 + h + h^2.$$

Thus

$$c_1(\mathcal{D}) = h, \quad c_2(\mathcal{D}) = h^2.$$

The signs depend on the convention for the tautological line, but the nontriviality of the bundle does not.

Equation (6) means that the dark-doublet connection cannot be globally removed by a single smooth choice of basis. Gauge patching in this example is therefore not an artifact of notation; it is forced by the topology of the protected relational bundle.

At the same time, the characteristic classes satisfy the special relation

$$c_2(\mathcal{D}) = c_1(\mathcal{D})^2.$$

This relation is not obeyed by a generic rank-two complex vector bundle. Consequently the bright-line model explores only a restricted subset of possible $U(2)$ bundle topology.

The limitation becomes especially clear on a compactified four-dimensional spacetime with the homotopy type of S^4 . Let

$$b : S^4 \longrightarrow \mathbb{CP}^2$$

be a bright-state texture and let

$$E = b^* \mathcal{D}$$

be the induced dark bundle. Since

$$H^2(S^4, \mathbb{Z}) = 0,$$

one has

$$b^* h = 0.$$

It follows that

$$c_1(E) = 0$$

and, using Eq. (6),

$$c_2(E) = b^*(h^2) = (b^* h)^2 = 0.$$

Proposition 12 (Topological limitation of the minimal $\mathbb{CP}^2$ model). A rank-two dark bundle obtained by pulling back the orthogonal-complement bundle

$$\mathcal{D} \rightarrow \mathbb{CP}^2$$

along a map

$$S^4 \rightarrow \mathbb{CP}^2$$

has vanishing second Chern number. The minimal bright-line construction therefore does not realize a nonzero $SU(2)$ instanton number on S^4 .

This result is useful precisely because the local curvature of the model is non-Abelian. Local non-Abelian holonomy and global gauge-bundle topology are distinct requirements. A construction may generate the full $\mathfrak{u}(2)$ Lie algebra locally while still being unable to realize all topological sectors of a general $U(2)$ or $SU(2)$ gauge theory.

The preceding examples suggest three logically separate levels of internal structure. The first is the kinematic operator algebra,

$$\mathcal{A}_{\text{int}}^{\text{kin}} = \text{End}(M),$$

which describes all linear transformations available on a multiplicity space. The second is the symmetry of the effective local dynamics,

$$G_{\text{sym}} \subseteq U(M),$$

which consists of transformations preserving the relevant Hamiltonian, constraints, and vacuum data. The third is the group actually generated by parallel comparison,

$$G_{\text{int}}^{\text{dyn}} = \overline{\text{Hol}(\nabla^M)} \subseteq G_{\text{sym}}.$$

These three objects need not coincide.

For example, a rank-two multiplicity space may support the complete algebra $M_2(\mathbb{C})$, while a Hamiltonian splitting

$$\Phi = \phi^a \sigma_a$$

selects a preferred relational direction and reduces the local symmetry to the stabilizer of ϕ . Conversely, the Hamiltonian may preserve the full degeneracy while the comparison curvature explores only one Abelian generator. The size of the observable algebra, the symmetry of the local spectrum, and the holonomy of transport are therefore distinct dynamical questions.

This distinction also clarifies the status of internal gauge emergence in the present construction. Sections 2–6 establish a route from projective many-body comparison to geometry-invisible quantum state spaces and from event-dependent protected multiplicities to non-Abelian internal holonomy. They do not yet establish that different localized matter sectors couple to one and the same internal connection. Without such a universality condition, separate multiplicity bundles could carry unrelated Berry or Wilczek–Zee connections even when their local holonomy groups are isomorphic.

A genuine internal interaction therefore requires an additional consistency principle linking the comparison transports of different sectors. The natural candidate is compatibility with composition and fusion: whether two relational excitations are fused before or after transport should not change the physical fusion process. This requirement constrains the connections on different multiplicity bundles simultaneously and leads naturally to a common tensor-automorphism group. We develop this comparison–fusion universality in the next section.

7 Fusion-Compatible Transport and the Reconstruction of a Common Internal Group

The preceding sections establish that a protected relational multiplicity may carry a noncommutative internal operator algebra and, when its embedding varies over spacetime, a nontrivial comparison connection with a dynamically selected holonomy group. This is still insufficient for a physical gauge interaction. Different localized sectors could, in principle, possess unrelated multiplicity bundles

$$\mathcal{M}_r, \quad \mathcal{M}_s, \quad \mathcal{M}_t, \dots$$

with unrelated connections

$$A^{(r)}, \quad A^{(s)}, \quad A^{(t)}, \dots$$

Even if several of these bundles have isomorphic fibers or isomorphic holonomy groups, this does not imply that they couple to one and the same internal comparison field.

The missing structure is supplied by composition. If relational sectors are physical localized excitations rather than isolated Hilbert-space factors, they must admit physically meaningful fusion channels. Let r , s , and t denote three such sectors and suppose that the fusion of r and s contains t . At an event x this channel is represented on the multiplicity spaces by an intertwiner

$$C_{rs}^t(x) : M_r(x) \otimes M_s(x) \longrightarrow M_t(x).$$

More generally, if the channel occurs with multiplicity greater than one, one may introduce an additional fusion label, but this refinement will not be needed for the structural argument below.

Let

$$U_\gamma^{(r)} : M_r(x) \longrightarrow M_r(y)$$

denote comparison transport of sector r along a path

$$\gamma : x \longrightarrow y.$$

There are then two operationally distinct descriptions of the same physical process. One may first fuse the two sectors at x and then transport the result,

$$M_r(x) \otimes M_s(x) \xrightarrow{C_{rs}^t(x)} M_t(x) \xrightarrow{U_\gamma^{(t)}} M_t(y),$$

or first transport the individual sectors and fuse them at y ,

$$M_r(x) \otimes M_s(x) \xrightarrow{U_\gamma^{(r)} \otimes U_\gamma^{(s)}} M_r(y) \otimes M_s(y) \xrightarrow{C_{rs}^t(y)} M_t(y).$$

If these two descriptions represent the same comparison process, the diagram must commute:

$$U_\gamma^{(t)} C_{rs}^t(x) = C_{rs}^t(y) (U_\gamma^{(r)} \otimes U_\gamma^{(s)}).$$

We shall refer to Eq. (7) as *fusion-compatible comparison transport*. It is stronger than requiring each individual multiplicity bundle to possess a unitary connection. It constrains the transports of different sectors simultaneously.

For infinitesimal transport, let

$$\nabla^{(r)} = d + A^{(r)}$$

denote the connection on $\mathcal{M}_r$. Applying Eq. (7) to an infinitesimal path gives the covariant constancy condition

$$\nabla C_{rs}^t = 0.$$

In local frames this reads

$$dC_{rs}^t + A^{(t)} C_{rs}^t - C_{rs}^t (A^{(r)} \otimes I + I \otimes A^{(s)}) = 0.$$

If the fusion tensor is constant in the chosen comparison frames,

$$dC_{rs}^t = 0,$$

then

$$A^{(t)} C_{rs}^t = C_{rs}^t (A^{(r)} \otimes I + I \otimes A^{(s)}).$$

The three connections are therefore not independent.

The same compatibility condition holds for curvature. Applying a second covariant derivative to Eq. (7) gives

$$F^{(t)} C_{rs}^t = C_{rs}^t (F^{(r)} \otimes I + I \otimes F^{(s)}).$$

Thus the curvature seen in a fused sector is constrained by the curvatures of its constituents. Fusion compatibility therefore acts not only on local frame choices but directly on the physically measurable holonomies.

This observation motivates the following principle.

Principle of comparison–fusion universality. The internal comparison transports of localized relational sectors are physical only if they preserve all admissible fusion, unit, and conjugation maps. Internal comparison is therefore a structure acting on the entire fusion system rather than an independent basis freedom attached separately to each sector.

To formulate this statement systematically, let

$$\mathcal{C}_{\text{rel}}$$

denote a category whose objects are localized relational sectors and whose morphisms are physical intertwiners among them. We assume that $\mathcal{C}_{\text{rel}}$ carries a tensor product

$$r \otimes s,$$

a unit object

$$\mathbf{1},$$

and, when required, dual or conjugate sectors

$$\bar{r}.$$

At each event x , assign to a sector r its relational multiplicity space

$$\omega_x(r) = M_r(x)$$

and to an intertwiner its corresponding linear map. This defines a fiber assignment

$$\omega_x : C_{\text{rel}} \longrightarrow \text{Hilb}_{\text{fd}}.$$

When tensor products and fusion maps are preserved, ω_x is a monoidal fiber functor in the usual representation-theoretic sense.

The comparison transport along a path $\gamma : x \rightarrow y$ then consists of a family

$$U_\gamma^{(r)} : \omega_x(r) \longrightarrow \omega_y(r)$$

for all sectors r . Equation (7) states precisely that this family is tensor-natural:

$$U_\gamma^{(r \otimes s)} = U_\gamma^{(r)} \otimes U_\gamma^{(s)}$$

after the appropriate fusion decomposition is taken into account, and

$$U_\gamma^{(t)} C = C U_\gamma^{(r \otimes s)}$$

for every physical intertwiner

$$C : r \otimes s \rightarrow t.$$

At a fixed event, consider the group of unitary tensor automorphisms of the fiber functor,

$$G_x := \text{Aut}^\otimes(\omega_x).$$

An element

$$g \in G_x$$

is a family of unitary maps

$$g_r : M_r(x) \longrightarrow M_r(x)$$

such that

$$g_{r \otimes s} = g_r \otimes g_s,$$

and

$$g_t C = C(g_r \otimes g_s)$$

for every admissible fusion intertwiner

$$C : M_r \otimes M_s \rightarrow M_t.$$

It also acts trivially on the tensor unit and compatibly on conjugate sectors.

Under the standard compactness, rigidity, and symmetry assumptions familiar from Tannaka–Krein reconstruction, the group $\text{Aut}^\otimes(\omega)$ reconstructs the compact group whose representations define the tensor category [?, ?, ?]. The abstract reconstruction theorem is not new. Its relevance here is that the candidate representation spaces arise as geometry-invisible relational multiplicities generated by composite projective comparison sectors, rather than being postulated as independent internal labels.

The reconstruction immediately explains why the unitary groups of the individual multiplicity spaces are too large. Before fusion constraints are imposed, the maximal independent basis freedom would be

$$\prod_r U(M_r).$$

Most elements of this product do not preserve the fusion tensors. The common internal group is instead the subgroup satisfying all conditions of the form Eq. (7):

$$G_{\text{int}} = \left\{ (g_r)_r \in \prod_r U(M_r) : g_t C_{rs}^t = C_{rs}^t (g_r \otimes g_s) \text{ for all fusion maps} \right\}.$$

Theorem 13 (Fusion reduction of independent multiplicity symmetries). Let $\{M_r\}$ be a family of finite-dimensional relational multiplicity spaces equipped with a specified set of tensor products, unit maps, duality maps, and fusion intertwiners. The unitary transformations that can act consistently on all sectors while leaving this relational composition structure invariant form the tensor-automorphism group

$$G_{\text{int}} = \text{Aut}^\otimes(\omega),$$

which is a subgroup of the product of independent multiplicity frame groups:

$$G_{\text{int}} \subseteq \prod_r U(M_r).$$

If comparison transport preserves all fusion data, its parallel transports take values in this common group rather than in independently chosen $U(M_r)$ factors.

The theorem changes the interpretation of internal universality. It is not necessary for all matter sectors to carry identical vector spaces or identical matrix-valued connections. What is required is that their transports arise from one common comparison structure through different representations.

Suppose, therefore, that the reconstructed compact group is G and that each multiplicity space carries a representation

$$\rho_r : G \longrightarrow U(M_r).$$

Let

$$P_G \longrightarrow M$$

be the corresponding principal comparison bundle and let

$$A \in \Omega^1(P_G, \mathfrak{g})$$

be a common principal connection. The multiplicity bundle of sector r is then the associated bundle

$$\mathcal{M}_r = P_G \times_{\rho_r} M_r,$$

and its induced connection is

$$A^{(r)} = \rho_{r*}(A).$$

Its curvature is correspondingly

$$F^{(r)} = \rho_{r*}(F).$$

This is the appropriate sense in which different relational sectors see the same internal comparison field. Their matrix representatives may differ, their dimensions may differ, and some sectors may be neutral, but the underlying principal connection is common.

Proposition 14 (Common-connection form of fusion-compatible transport). Assume that the relational sector category is equivalent to a full tensor subcategory of finite-dimensional unitary representations of a compact group G , and that comparison transport is tensor-natural. Then locally there exists a common $\mathfrak{g}$ -valued connection A such that every sector connection can be written as

$$A^{(r)} = \rho_{r*}(A).$$

Fusion compatibility is then automatic because every fusion tensor is a G -intertwiner.

The proposition is a reconstruction statement rather than a derivation of a particular group. The actual group depends on the relational fusion data. The physical content is that gauge universality should be sought in the compatibility of comparison with fusion, not in the accidental equality of Berry connections defined separately in different sectors.

The Abelian case gives a particularly transparent consequence. Let

$$G = U(1)$$

and let sector r transform with charge q_r :

$$\rho_r(e^{i\theta}) = e^{iq_r\theta}.$$

If a fusion channel

$$C_{rs}^t \neq 0$$

exists, invariance of the intertwiner gives

$$e^{iq_t\theta} C_{rs}^t = e^{i(q_r+q_s)\theta} C_{rs}^t.$$

Since the intertwiner is nonzero,

$$q_t = q_r + q_s.$$

Corollary 15 (Charge additivity from fusion compatibility). For a reconstructed $U(1)$ internal comparison group, every allowed fusion channel

$$r \otimes s \rightarrow t$$

satisfies

$$q_t = q_r + q_s.$$

Thus Abelian charge conservation is equivalent to invariance of the fusion map under the common internal comparison transformation.

The same argument determines the charge of the conjugate sector. If

$$r \otimes \bar{r} \rightarrow \mathbf{1}$$

contains the tensor unit and the vacuum is neutral,

$$q_1 = 0,$$

then

$$q_{\bar{r}} = -q_r.$$

In this formulation, the sign reversal of the conjugate charge is a consequence of the existence of an invariant evaluation map rather than an independent rule.

For a non-Abelian group the corresponding statement is representation theoretic rather than additive. A fusion channel

$$r \otimes s \rightarrow t$$

is allowed only when

$$\text{Hom}_G(M_r \otimes M_s, M_t) \neq 0.$$

The familiar Clebsch–Gordan selection rules are therefore the non-Abelian counterpart of Eq. (7). In the present interpretation, they express the condition that the relational composition law remain meaningful under internal comparison.

The same structure constrains interaction vertices. Let

$$\Phi_r, \quad \Phi_s, \quad \Phi_t$$

denote effective fields associated with three relational sectors. A local interaction of the form

$$\mathcal{L}_{\text{int}} \supset \lambda C_{rs}^t \Phi_t^\dagger \Phi_r \Phi_s + \text{h.c.}$$

is invariant under the common internal comparison group precisely when C_{rs}^t is an intertwiner. The interaction selection rule and the transport compatibility condition are therefore controlled by the same fusion tensor.

This gives a more precise meaning to the statement that internal charge is a response property. The charge of a sector is not interpreted as an additional geometric coordinate carried by a localized object. It specifies how the relational multiplicity transforms when the common internal comparison frame is changed:

$$\delta_\epsilon \psi_r = -\rho_{r*}(\epsilon) \psi_r.$$

Different charges or non-Abelian representations correspond to different responses to the same underlying comparison generator.

There is a close structural parallel with geometric comparison. A localized matter carrier with spacetime spin transforms under a geometric representation of the Lorentz or spin group, while its multiplicity factor transforms under the reconstructed internal group. Schematically,

$$\mathcal{E}_r = S_r \otimes M_r,$$

where S_r denotes the geometric spinorial or tensorial carrier and M_r the relational multiplicity. If the two comparison structures act on the two factors separately, the covariant derivative takes the form

$$\nabla^{(r)} = \nabla_{S_r}^{\text{geom}} \otimes I + I \otimes \nabla_{M_r}^{\text{int}}.$$

For a spinorial carrier this becomes locally

$$\nabla_\mu^{(r)} = \partial_\mu + \frac{1}{4} \omega_{\mu ab} \gamma^{ab} + \rho_{r*}(A_\mu).$$

The geometric and internal connections are therefore distinguished by the structures they compare: the first acts on the spacetime carrier, while the second acts on geometry-invisible relational multiplicity.

Their curvatures similarly decompose,

$$[\nabla_\mu^{(r)}, \nabla_\nu^{(r)}] = \frac{1}{4} R_{\mu\nu ab} \gamma^{ab} + \rho_{r*}(F_{\mu\nu}),$$

provided the two actions commute. This does not identify gravitational and internal curvature. It places them as two components of the complete comparison curvature experienced by a matter carrier.

The existence of the split form Eq. (7) also clarifies why an internal connection can change a matter state without altering its local spacetime geometry. The internal term belongs to the commutant of the geometric action:

$$[\rho_{r*}(A_\mu), \gamma^a] = 0$$

when the multiplicity acts on a separate tensor factor. Internal comparison therefore preserves the local Clifford relations while rotating the geometry-invisible relational state.

This local tensor-product separation need not imply a globally direct-product structure group. If the spin cover and the internal group share central elements that act identically on the matter carrier, the true global structure group may be a quotient

$$\frac{\text{Spin}(1, 3) \times G}{\Gamma_{\text{center}}}.$$

Such global identifications are relevant to Spin^G and Spin^c structures, but their analysis requires information about the topology and center action of the reconstructed internal group and will not be assumed in the local development below.

The category-theoretic formulation also makes clear what has not yet been derived. A tensor-automorphism group can be reconstructed only after the relevant categorical data exist. The MS comparison programme must still establish, for any proposed physical realization, that the projective relational sectors are sufficiently localized and transportable, that their fusion is associative in the appropriate sense, that conjugate sectors and evaluation maps exist when required, and that the statistics and locality structure satisfy the hypotheses appropriate to the reconstruction theorem being used. In relativistic algebraic quantum field theory these inputs are substantial, and the Doplicher–Haag–Roberts and Doplicher–Roberts reconstruction theorems should be regarded as conditional targets rather than automatic consequences of projective spin fractionalization [?, ?, ?].

There is also a dynamical issue not addressed by Eq. (7). The reconstruction determines the allowed common internal comparison group and the representation content of its sectors, but it does not determine whether the corresponding connection is an independently propagating field. A tensor-natural Berry or Wilczek–Zee connection may remain completely slaved to slowly varying relational order parameters. In that case the structure derived above is a genuine internal bundle and comparison law, but not yet a Yang–Mills dynamics.

The logical status of the construction may therefore be summarized by the chain

projective relational sectors $\longrightarrow$ multiplicity spaces $\longrightarrow$ fusion data $\longrightarrow \text{Aut}^\otimes(\omega) \longrightarrow$ common internal comparison

where each arrow introduces a stronger physical requirement. The final step from a common internal comparison connection to a propagating gauge field is dynamical rather than purely representation theoretic. It requires the connection to acquire an independent low-energy response kernel and a finite positive curvature stiffness. We turn to this connection-liberation problem next.

8 Connection Liberation and the Dynamical Gauge Criterion

The previous section identifies the structural conditions under which different relational sectors can be acted on by one common internal comparison connection. This result should not be confused with the emergence of a dynamical gauge field. A connection may be induced entirely by the motion of a finite set of relational order parameters and may possess nontrivial holonomy without representing an independent low-energy degree of freedom. The final step from internal comparison geometry to Yang–Mills dynamics is therefore a dynamical one.

To make this distinction precise, consider first an internal connection that is completely induced by a family of slow relational parameters

$$\lambda^A(x), \quad A = 1, \dots, n_\lambda.$$

Suppose that a protected multiplicity bundle over the parameter space carries a Berry or Wilczek–Zee connection

$$\mathcal{A} = \mathcal{A}_A(\lambda) d\lambda^A.$$

After the parameter field is promoted to a spacetime-dependent configuration, the induced spacetime connection is

$$A_\mu(x) = \mathcal{A}_A(\lambda(x)) \partial_\mu \lambda^A(x).$$

Its curvature is the pullback of the parameter-space curvature,

$$F_{\mu\nu}(x) = \mathcal{F}_{AB}(\lambda(x)) \partial_\mu \lambda^A \partial_\nu \lambda^B,$$

where

$$\mathcal{F}_{AB} = \partial_A \mathcal{A}_B - \partial_B \mathcal{A}_A + [\mathcal{A}_A, \mathcal{A}_B].$$

Equations (8) and (8) describe a genuine connection and may produce non-Abelian holonomy. Nevertheless, the spacetime connection is completely slaved to the scalar fields λ^A . Its local variations are restricted to those obtained from variations of the map

$$\lambda : M \longrightarrow \mathcal{P},$$

where $\mathcal{P}$ is the parameter manifold of protected relational subspaces.

This gives a first obstruction.

Proposition 16 (Finite-parameter pullback obstruction). Let an internal connection on spacetime be of the form

$$A_\mu = \mathcal{A}_A(\lambda) \partial_\mu \lambda^A$$

for a finite-dimensional parameter field $\lambda : M \rightarrow \mathcal{P}$. Then its curvature satisfies

$$F_{\mu\nu} = \mathcal{F}_{AB}(\lambda) \partial_\mu \lambda^A \partial_\nu \lambda^B.$$

Consequently the connection configurations belong to the pullback image of the parameter-space connection and do not constitute an independent generic Yang–Mills configuration space.

The proposition does not imply that a composite connection is physically irrelevant. Composite gauge fields can mediate nontrivial holonomy and can become useful low-energy variables. It states

only that the existence of a Berry or Wilczek–Zee connection is insufficient to establish an independent gauge degree of freedom.

There is also a derivative-order distinction. If the fundamental slow fields have an ordinary sigma-model kinetic term,

$$S_\lambda = \frac{f^2}{2} \int d^4x G_{AB}(\lambda) \partial_\mu \lambda^A \partial^\mu \lambda^B,$$

then

$$F_{\mu\nu} = O((\partial\lambda)^2),$$

and therefore

$$\text{tr } F_{\mu\nu} F^{\mu\nu} = O((\partial\lambda)^4).$$

A Yang–Mills term constructed from the composite curvature is thus a higher-derivative operator from the viewpoint of the underlying scalar description. This is different from an effective theory in which A_μ itself is an independent low-energy variable and

$$F_{\mu\nu} F^{\mu\nu}$$

is its leading kinetic term.

A convenient way to formulate the required enlargement is to return to the Stiefel-frame description of the relational multiplicity bundle. Let

$$Z^\dagger Z = I_m$$

and introduce an auxiliary matrix-valued connection A_μ through

$$D_\mu Z = \partial_\mu Z - Z A_\mu.$$

Consider an effective action of the schematic form

$$S_{\text{St}} = f^2 \int d^4x \text{tr} [(D_\mu Z)^\dagger D^\mu Z] + S_{\text{gap}}[Z] + \dots$$

The action is invariant under

$$Z(x) \mapsto Z(x)U(x),$$

provided

$$A_\mu \mapsto U^{-1} A_\mu U + U^{-1} \partial_\mu U.$$

If no independent kinetic term for A_μ is present, variation of Eq. (8) with respect to A_μ gives an algebraic equation. In the simplest normalization it yields

$$A_\mu = Z^\dagger \partial_\mu Z,$$

which is precisely the induced multiplicity connection. At tree level the connection therefore carries no independent propagating degree of freedom.

Definition. We shall say that *connection liberation* occurs when an internal comparison connection that is auxiliary or composite in the microscopic description becomes an independent low-energy collective field whose effective action contains a nondegenerate kinetic operator not fixed algebraically by the instantaneous relational projector.

This definition separates two logically different questions:

Does a relational connection exist?

and

Does that connection propagate independently?

Sections 5–7 address the first question. The remainder of the present section concerns the second.

A standard route to connection liberation is quantum or statistical polarization of gapped microscopic degrees of freedom [?, ?]. Let χ collectively denote relational excitations coupled covariantly to the internal comparison connection. The effective action obtained by integrating them out is defined by

$$e^{i\Gamma[A]} = \int \mathcal{D}\chi e^{iS_{\text{rel}}[\chi, A]}.$$

Expanding about a symmetric comparison vacuum gives

$$\Gamma[A] = \Gamma[0] + \Gamma^{(1)}[A] + \Gamma^{(2)}[A] + \cdots.$$

For a vacuum carrying no internal current,

$$\Gamma^{(1)}[A] = 0.$$

The quadratic term can be written

$$\Gamma^{(2)} = \frac{1}{2} \int \frac{d^4k}{(2\pi)^4} A_\mu^a(-k) \Pi^{ab\mu\nu}(k) A_\nu^b(k).$$

The kernel

$$\Pi_{\mu\nu}^{ab}(k)$$

is the internal comparison response function. When A_μ couples to a conserved internal current J_μ^a , it is determined, up to local contact terms, by the connected current correlation function

$$\Pi_{\mu\nu}^{ab}(x-y) \sim i \left\langle J_\mu^a(x) J_\nu^b(y) \right\rangle_c.$$

Thus the kinetic properties of the emergent comparison connection are controlled by fluctuations of the same relational degrees of freedom whose fusion structure determines the internal group.

If the internal comparison symmetry is unbroken and realized consistently at the quantum level, the corresponding Ward identity imposes

$$k^\mu \Pi_{\mu\nu}^{ab}(k) = 0.$$

In a Lorentz-invariant vacuum and for a simple isotropic internal sector, the leading transverse structure has the form

$$\Pi_{\mu\nu}^{ab}(k) = \delta^{ab} (k^2 \eta_{\mu\nu} - k_\mu k_\nu) \Pi(k^2) + \cdots,$$

where the ellipsis allows group-dependent invariant tensors and higher-derivative terms in more general cases.

Suppose the relational excitations integrated out in Eq. (8) are gapped at a scale Δ . At momenta

$$|k| \ll \Delta,$$

locality of the infrared expansion permits

$$\Pi(k^2) = K_A + O\left(\frac{k^2}{\Delta^2}\right).$$

The effective action then contains

$$\Gamma_{\text{eff}}[A] \supset -\frac{K_A}{4} \int d^4x F_{\mu\nu}^a F^{a\mu\nu} + \dots$$

Once the Lie-algebra normalization is fixed, it is natural to define

$$g_{\text{eff}}^{-2} := K_A.$$

The coefficient K_A has a direct response interpretation. A large positive K_A penalizes internal comparison curvature strongly, whereas a small K_A corresponds to a more compliant internal comparison sector. We shall therefore call

$$K_A$$

the *internal comparison stiffness*. Its inverse is the corresponding long-wavelength comparison susceptibility.

This interpretation parallels the constitutive role of Newton's constant in the gravitational sector of the MS programme. There the operational soldering data determine the geometric area variable, while the coefficient multiplying the Hodge-dual gravitational excitation remains an independent constitutive stiffness [?]. The present construction suggests the analogous schematic correspondence

$$G^{-1} \leftrightarrow \text{geometric comparison stiffness},$$

and

$$g_{\text{eff}}^{-2} \leftrightarrow \text{internal comparison stiffness}.$$

The analogy concerns response structure, not numerical equality. The two coefficients carry different dimensions in four spacetime dimensions and are determined by different tensor channels of the microscopic response theory.

The appearance of a curvature term is not by itself sufficient for a massless gauge field. The most direct diagnostic is the infrared two-point function of the liberated connection. After gauge fixing, write the transverse part of the one-particle-irreducible quadratic kernel as

$$\Gamma_{T,\mu\nu}^{(2)ab}(k) = \delta^{ab} \left[m_A^2 + K_A k^2 + O\left(\frac{k^4}{\Delta^2}\right) \right] P_{\mu\nu}^T(k),$$

where

$$P_{\mu\nu}^T(k) = \eta_{\mu\nu} - \frac{k_\mu k_\nu}{k^2}.$$

For an exactly unbroken local gauge redundancy the mass term is forbidden unless additional Higgs or Stueckelberg structure is present. In an emergent description, however, the vanishing of m_A^2 is a nontrivial infrared requirement that must be protected by the reconstructed comparison symmetry or established dynamically.

Definition. A relational internal connection will be called a *massless liberated comparison field* when its infrared transverse kernel satisfies

$$m_A^2 = 0$$

and

$$0 < K_A < \infty.$$

Under these conditions the transverse propagator has the leading form

$$\langle A_\mu^a(k) A_\nu^b(-k) \rangle \simeq \frac{\delta^{ab}}{K_A} \frac{P_{\mu\nu}^T(k)}{k^2 + i0},$$

or, using Eq. (8),

$$\langle A_\mu^a(k) A_\nu^b(-k) \rangle \simeq g_{\text{eff}}^2 \delta^{ab} \frac{P_{\mu\nu}^T(k)}{k^2 + i0}.$$

Theorem 17 (Infrared gauge-field criterion). Assume that a common fusion-compatible internal comparison connection has been reconstructed and that the low-energy effective theory admits a local one-particle-irreducible expansion around an unbroken comparison-symmetric vacuum. The connection describes an independently propagating massless gauge mode only if its transverse inverse propagator possesses a simple zero at

$$k^2 = 0$$

with positive finite residue. In the expansion Eq. (8), this requires

$$m_A^2 = 0, \quad 0 < K_A < \infty.$$

A nonzero Berry curvature without these conditions does not establish a massless gauge field.

The theorem is best understood as a diagnostic rather than a microscopic derivation. It identifies the observable infrared property that any proposed MS comparison dynamics must reproduce if the internal connection is to be interpreted as a Yang–Mills field.

There are several distinct ways in which connection liberation can fail. If

$$K_A = 0,$$

no leading quadratic curvature stiffness is generated, and the connection remains nondynamical at this order. If

$$K_A < 0,$$

the putative vacuum is unstable or the assumed low-energy truncation is inconsistent. If

$$m_A^2 > 0,$$

the connection has no long-range massless pole unless the mass belongs to a consistent Higgs or Stueckelberg phase. Finally, if the effective action contains only the pullback structure of Eq. (8), then the connection remains a constrained collective coordinate even when its curvature is nonzero.

The distinction between a connection and an internal order parameter should also be retained. Suppose the protected multiplicity sector carries a traceless splitting

$$\Phi(x) = \phi^a(x) T_a.$$

A nonzero expectation value

$$\langle \Phi \rangle \neq 0$$

selects preferred directions in the multiplicity space and reduces the symmetry to the stabilizer of the order parameter. This is a symmetry-breaking or Higgs-like structure. It is not itself a connection. Conversely, a nontrivial A_μ can exist while

$$\Phi = 0$$

and the local multiplicity spectrum remains exactly degenerate.

Thus at least four structures must be distinguished:

$$\text{End}(M), \quad G_{\text{sym}}, \quad A_\mu, \quad \Phi.$$

They represent, respectively, the kinematic internal operator algebra, the unbroken symmetry of the relational dynamics, the comparison connection, and an internal symmetry-breaking order parameter.

The relation between protection and liberation is particularly important. The multiplicity space is useful as an internal carrier only if it is separated from unwanted sectors strongly enough to remain well defined at low energy. Let

$$\Delta_{\text{rel}} > 0$$

denote such a protective gap. At the same time, a long-range gauge field requires the infrared obstruction to propagation to vanish:

$$m_A^2 = 0.$$

The two requirements are logically independent:

$$\Delta_{\text{rel}} > 0$$

protects the relational carrier, whereas

$$m_A^2 = 0$$

liberates its comparison connection.

This motivates the following structural principle.

Protection–liberation principle. An emergent internal gauge sector requires both a protected relational multiplicity and an infrared-liberated comparison connection. A positive gap may stabilize the internal carrier without producing a long-range gauge field, while a massless collective mode without a protected internal sector need not define a stable relational charge.

The same logical pattern appears in projective comparison-string models. A gap can protect the projective edge class while deconfinement requires a separate vanishing of the infrared string tension. In the present setting, the corresponding pair of conditions is

$$\Delta_{\text{rel}} > 0, \quad m_A^2 = 0.$$

The recurrence of this structure suggests that protected information and liberated transport should be treated as separate dynamical questions throughout the MS quantum-comparison programme.

The common principal connection reconstructed in Section 7 also constrains the interpretation of the coupling constant. Once the normalization of the generators $T_a \in \mathfrak{g}$ is fixed, the coefficient

$$g_{\text{eff}}$$

belongs to the gauge sector rather than independently to each matter representation. Different sectors respond through

$$\rho_{r*}(T_a),$$

while the common kinetic term is

$$S_{\text{YM}} = -\frac{1}{4g_{\text{eff}}^2} \int d^4x \kappa_{ab} F_{\mu\nu}^a F^{b\mu\nu}.$$

Representation-dependent charges and one common curvature stiffness are therefore different pieces of data.

This distinction is clearest for a $U(1)$ factor. The covariant derivative of sector r has the form

$$D_\mu^{(r)} = \partial_\mu + iq_r A_\mu,$$

while the field strength term is normalized once:

$$S_{U(1)} = -\frac{1}{4g^2} \int d^4x F_{\mu\nu} F^{\mu\nu}.$$

The fusion rule determines the additive representation labels

$$q_t = q_r + q_s,$$

whereas the vacuum polarization of the common comparison field determines g^{-2} . Charge and stiffness therefore have distinct structural origins.

The effective action also provides a natural response definition of the internal current. For sector r ,

$$J_a^\mu = \frac{\delta S_{\text{matter}}}{\delta A_\mu^a}.$$

The current measures the response of the relational state to an infinitesimal deformation of the internal comparison connection. The connected two-current kernel then measures the response of the comparison vacuum itself:

$$\langle J_a^\mu(x) J_b^\nu(y) \rangle_c \longrightarrow K_A.$$

In this sense, an internal charge specifies how a relational excitation responds to a comparison generator, while the gauge coupling specifies how stiffly the comparison vacuum responds to curvature sourced by those excitations.

This response interpretation produces a close but nonidentical parallel with gravity. In the gravitational MS construction, the soldering response first determines

$$e, \quad g, \quad B = e \wedge e,$$

whereas the constitutive law determines how strongly the gravitational excitation responds to the geometric area. The overall coefficient remains independent of the reconstructed mean geometry and is interpreted as gravitational stiffness [?]. In the present internal construction, representation and fusion data determine the comparison group and its action, while the quantum response kernel determines the curvature stiffness g_{eff}^{-2} .

The resulting parallel is therefore

gravitational comparison	internal comparison
soldered geometric carrier	relational multiplicity carrier
ω	A
$R(\omega)$	$F(A)$
G^{-1}	g_{eff}^{-2}
geometric response stiffness	internal response stiffness

where the table is structural rather than numerical. No equality or fixed relation between G and g_{eff} follows from the construction.

The combined result of the preceding sections can now be stated more sharply. A relational multiplicity becomes a candidate internal gauge sector only after several logically distinct steps:

- projective composite carriers
 - geometry-invisible multiplicity
 - noncommutative relational algebra
 - protected multiplicity bundle
 - nontrivial comparison holonomy
 - fusion-compatible common connection
 - independent massless transverse mode.

Only the final structure deserves the interpretation of a propagating Yang–Mills field.

The present paper does not derive Eq. (8) from a complete microscopic MS Hamiltonian. In particular, it does not calculate K_A , prove the vanishing of m_A^2 , or establish the existence of a deconfined comparison phase in which the liberated connection survives as an autonomous long-distance degree of freedom. These are dynamical targets for a microscopic completion. What the analysis provides is a separation theorem in practice: multiplicity, non-Abelian holonomy, common gauge reconstruction, and gauge-boson propagation are different layers and must not be inferred from one another without the corresponding dynamical input.

This separation will be important in assessing the scope of the proposed mechanism. The relational construction supplies a possible origin for internal quantum state spaces and for the geometric form of gauge comparison, but it does not yet select the Standard Model gauge group, its chiral matter representations, anomaly cancellation conditions, or its numerical couplings. We collect these limitations and the main structural results in the final discussion.

9 Discussion and Scope

The purpose of the present work has been to isolate a possible origin of internal quantum structure from relational multiplicity rather than to postulate an internal gauge group at the outset. The central distinction is between the geometric representation carried by a composite projective sector and the multiplicity with which that representation occurs. Once a collective geometric representation V_j has been fixed, the remaining factor

$$M_j$$

contains coherent information that is unresolved by the corresponding geometric comparison action. The main question addressed here is how such geometry-invisible multiplicity can acquire the algebraic, geometric, and dynamical structures normally associated with an internal gauge sector.

The resulting construction is deliberately layered. The first layer is finite-dimensional and representation theoretic:

$$\left(V_{\frac{1}{2}}\right)^{\otimes N} = \bigoplus_j V_j \otimes M_j.$$

The second is algebraic: rotationally invariant pair relations act noncommutatively on the multiplicity factors and, in the minimal examples, generate complete matrix algebras. The third is geometric: protected event-dependent multiplicities define vector bundles and induced comparison connections. The fourth is group theoretic: curvature and holonomy select a dynamically realized subgroup of the maximal multiplicity frame group. The fifth is compositional: fusion-compatible transport constrains otherwise independent sector connections and permits the reconstruction of a common internal comparison group. The final layer is dynamical: the reconstructed connection must become an independent massless transverse mode before it can be interpreted as a Yang–Mills field.

Keeping these layers separate is essential. Several implications that may appear natural at an intuitive level are not valid:

$$\dim M = m \not\Rightarrow G_{\text{int}} = U(m),$$

$$F \neq 0 \not\Rightarrow \text{propagating gauge boson},$$

and

$$\text{Aut}^{\otimes}(\omega) \not\Rightarrow \text{a specific microscopic Yang–Mills dynamics}.$$

The construction developed in this paper is intended precisely to identify the additional conditions required at each of these transitions.

What is established without a microscopic gauge assumption. Several results do not depend on postulating a conventional internal gauge field. Once a minimal spin- $\frac{1}{2}$ projective carrier is taken as input, its tensor powers necessarily contain multiplicity spaces. The four-carrier singlet sector

$$\mathcal{I}_4 \simeq \mathbb{C}^2$$

provides the simplest completely geometry-neutral example. Pair-neutrality projectors restricted to this space do not commute and generate

$$\text{End}(\mathcal{I}_4) \simeq M_2(\mathbb{C}).$$

The internal state space is therefore quantum in the operator-algebraic sense; it cannot be modeled as a set of simultaneously definite classical pairing labels.

The same neutral sector is invisible to one-carrier rotational tomography:

$$\rho_i = \frac{1}{2}I.$$

Moreover,

$$P_{\text{inv}} J_i^a P_{\text{inv}} = 0,$$

so a local vector perturbation possesses no first-order matrix element within the total-spin-zero subspace. These statements provide a more precise criterion of internality than collective invariance

alone. The information is absent from every individual carrier and is accessed only through multi-carrier relational observables.

The capacity result further shows that this is not a small finite-system effect. For $N = 2n$,

$$\dim \mathcal{I}_N = C_n \sim \frac{4^n}{\sqrt{\pi} n^{3/2}},$$

and hence

$$\log_2 \dim \mathcal{I}_N = N - \frac{3}{2} \log_2 N + O(1).$$

The fraction of the original tensor-product information capacity lost by imposing total rotational neutrality vanishes asymptotically. A macroscopically large relational system can therefore remain externally simple while retaining an extensive internal quantum state space.

These finite-dimensional facts are standard consequences of spin representation theory and its commutant structure. The proposed contribution is their interpretation inside the MS reconstruction: the multiplicity factor is identified as the coherent residue left after geometric comparison has fixed the representation carried by the composite excitation.

Why the commutant should not be identified with a gauge group. The geometric commutant

$$\mathcal{A}'_{\text{geom}} \simeq \bigoplus_j I_{V_j} \otimes \text{End}(M_j)$$

specifies the maximal algebra of operators invisible to the collective geometric action. It does not determine which operators occur dynamically or which transformations are physically transportable between events.

This point is particularly important because a multiplicity factor $M \simeq \mathbb{C}^m$ always admits a kinematic frame group $U(m)$. If this frame group were identified directly with the physical gauge group, the construction would merely rename a basis freedom as an interaction. The fixed-projector result of Section 5 provides a simple diagnostic:

$$P(x) = P_0, \quad F_D = 0 \quad \implies \quad F_A = 0.$$

A changing basis inside one fixed physical subspace produces a pure-gauge connection. Nontrivial relational curvature requires the physical projector itself to vary or an independent ambient curvature to be present.

The appropriate physical group is therefore the holonomy group

$$G_{\text{int}}^{\text{dyn}} = \overline{\text{Hol}(\nabla^{\mathcal{M}})},$$

whose Lie algebra is generated by transported curvature values. In particular, a rank-two multiplicity supports the full matrix algebra $M_2(\mathbb{C})$ but can possess only Abelian connected holonomy if all curvature values lie in one commuting direction. The appearance of two noncommuting traceless curvature directions is the actual local criterion for the generation of $\mathfrak{su}(2)$.

The $\mathbb{CP}^2$ dark-doublet example illustrates both the power and the limitation of this statement. A moving complex two-plane inside $\mathbb{C}^3$ can generate generic $\mathfrak{u}(2)$ local holonomy, whereas a restriction to a real comparison family reduces the connected rank-two frame group to $SO(2)$. At the same time, the corresponding dark bundle satisfies the special relation

$$c_2 = c_1^2,$$

so its local non-Abelianity does not imply access to all global $U(2)$ bundle topologies. In particular, its pullback to S^4 cannot carry a nonzero second Chern number. Local Lie-algebra generation and global topological universality must therefore be tested separately.

Fusion as the source of gauge universality. The existence of non-Abelian holonomy in one relational bundle still does not explain why different matter sectors should couple to the same gauge field. This motivates the comparison–fusion principle developed in Section 7. If a physical fusion channel

$$C_{rs}^t : M_r \otimes M_s \rightarrow M_t$$

is meaningful independently of the event at which it is evaluated, comparison transport must satisfy

$$U_\gamma^{(t)} C_{rs}^t = C_{rs}^t (U_\gamma^{(r)} \otimes U_\gamma^{(s)}).$$

This requirement turns a collection of independent vector-bundle transports into a tensor-natural family.

The common internal group is then not the product

$$\prod_r U(M_r)$$

but the subgroup preserving the complete fusion system. Under the usual hypotheses of compact tensor reconstruction this group is

$$G_{\text{int}} = \text{Aut}^\otimes(\omega).$$

The corresponding sector connections take the associated-bundle form

$$A^{(r)} = \rho_{r*}(A).$$

Gauge universality is therefore expressed as the existence of one comparison connection acting through different representations, rather than as equality of matrix-valued connections of different dimensions.

In the Abelian case this immediately yields

$$r \otimes s \rightarrow t \implies q_t = q_r + q_s.$$

Charge conservation can therefore be read as invariance of relational fusion under internal comparison. In the non-Abelian case the analogous statement is the existence of invariant Clebsch–Gordan intertwiners. This does not derive the observed charge spectrum; it identifies the structural origin of charge additivity once the internal comparison group and its sector content have been reconstructed.

The use of Tannaka–Krein or Doplicher–Roberts reconstruction at this stage must remain conditional. The mathematical theorems require appropriate tensor, rigidity, conjugation, locality, and transportability properties. Projective comparison fractionalization alone does not establish those properties. A microscopic MS completion must therefore show that the candidate relational excitations form a sufficiently well behaved category before the reconstruction theorem can be applied physically.

The dynamical gap between holonomy and Yang–Mills theory. The final distinction made in this paper concerns the status of the connection itself. A Berry or Wilczek–Zee connection may possess nonzero curvature and non-Abelian holonomy while remaining completely determined by a finite set of slowly varying microscopic parameters:

$$A_\mu = \mathcal{A}_A(\lambda) \partial_\mu \lambda^A.$$

Such a field is composite. It does not furnish a generic independent Yang–Mills configuration.

Connection liberation requires an additional infrared phenomenon. The internal comparison field must acquire a transverse quadratic kernel of the form

$$\Gamma_T^{(2)}(k) = [m_A^2 + K_A k^2 + O(k^4)] P_T.$$

A massless propagating comparison field requires

$$m_A^2 = 0, \quad 0 < K_A < \infty.$$

The coefficient

$$K_A = g_{\text{eff}}^{-2}$$

then plays the role of an internal comparison stiffness. Neither its positivity nor its magnitude follows from multiplicity counting, Grassmannian geometry, holonomy, or tensor reconstruction. It must be calculated from the microscopic comparison dynamics.

This separation is analogous to the treatment of gravitational stiffness in the MS gravitational sector. The operational response reconstructs the geometric arena, but does not fix Newton’s constant. The constitutive coefficient multiplying the gravitational response must be supplied by additional dynamics [?]. In the present case, the relational representation and fusion structure may reconstruct the internal comparison group while the vacuum response kernel determines the gauge stiffness.

The analogy suggests

$$G^{-1} \sim \text{geometric comparison stiffness}, \quad g^{-2} \sim \text{internal comparison stiffness},$$

but no numerical relation between the two is implied. They belong to different response channels and have different dimensional character in four spacetime dimensions.

Relation to the broader MS quantum-comparison programme. The present construction also clarifies the role of internal structure relative to the quantum geometry problem. The exact local MS response provides a commuting configuration algebra under exact additivity, so noncommutative quantum structure cannot be obtained merely by broadening a probability distribution over the same soldering spectrum [?]. Projective comparison and relational multiplicity belong to precisely the enlarged sector needed to escape that classicality result.

The internal noncommutativity found here should nevertheless not be confused with noncommutativity of spacetime geometry. The operators

$$P_{ij}^{(0)}$$

act within a multiplicity space while preserving the collective geometric representation. Their failure to commute demonstrates the quantum character of relational organization, not a failure of space-time coordinates or soldering variables to commute. In this sense the present mechanism supplies

a quantum internal sector without requiring the local geometric response algebra itself to become noncommutative.

There is also a sharp distinction between the projective comparison string that can generate a half-integer endpoint and the geometric response loop used to probe torsion. The former is a many-body comparison structure whose boundary can carry a nontrivial spin-cover class; the latter is constructed from tensorial geometric variables and probes closed displacement response. They should not be identified. The present paper begins only after the projective sector is assumed to exist and asks what new internal structure its many-body composition permits.

What remains open. The present construction does not derive the Standard Model. In particular, it does not determine

$$SU(3) \times SU(2) \times U(1),$$

the observed chiral representations, the number of generations, hypercharge assignments, the Higgs sector, the numerical gauge couplings, or the observed hierarchy of masses. It also does not establish relativistic Fermi statistics from the projective invariant $\Xi = -I$. The latter identifies a half-integer projective rotational class, whereas the spin–statistics theorem requires additional locality, covariance, spectral, and positivity assumptions.

Nor does the present construction establish anomaly cancellation. If a future MS completion generates chiral matter sectors, preservation of the quantum comparison symmetry would require the corresponding gauge and mixed gravitational anomalies to cancel. Such conditions would constitute further consistency constraints on the reconstructed fusion and representation system, not consequences of multiplicity alone.

A complete microscopic theory would therefore have to solve at least the following chain of dynamical problems:

- production of localized projective sectors,
- selection of a stable low-energy multiplicity structure,
- deconfinement and transportability of the sectors,
- derivation of their fusion and conjugation rules,
- selection of the physical holonomy group,
- liberation of a massless comparison connection,
- derivation of its kinetic stiffness,
- and quantum consistency of the resulting chiral theory.

The contribution of the present work is not to remove all of these steps but to place them in a definite logical order.

The main conceptual proposal may be stated more compactly. Internal quantum degrees of freedom need not be added as primitive labels attached to otherwise complete geometric particles. They can instead be sought in the coherent multiplicity of relational realizations of one and the same geometric representation. Gauge structure then concerns the consistent comparison of those multiplicities across spacetime and across fusion processes. Whether this kinematic and categorical structure is promoted to a dynamical Yang–Mills field is a final infrared question rather than an automatic consequence of internal state counting.

10 Conclusion

We have developed a route from projective quantum comparison to internal gauge structure within the motion–staticity reconstruction programme. The starting point is a minimal projective carrier

$$V_{\frac{1}{2}} \simeq \mathbb{C}^2,$$

taken as an infrared input from an enlarged comparison sector. When several such carriers are composed, their collective rotational representation decomposes as

$$\left(V_{\frac{1}{2}}\right)^{\otimes N} = \bigoplus_j V_j \otimes M_j.$$

The factors M_j are invisible to the collective geometric action and provide the basic relational multiplicity spaces studied in this work.

These multiplicities are not merely classical degeneracy labels. In the four-carrier singlet sector,

$$\mathcal{I}_4 \simeq \mathbb{C}^2,$$

rotationally invariant pair-neutrality projectors fail to commute and generate the complete algebra

$$M_2(\mathbb{C}).$$

The resulting qubit is entirely relational: every individual constituent is maximally mixed, and local vector operators possess no first-order matrix elements within the neutral sector. More generally, the neutral state space of $N = 2n$ carriers has dimension

$$C_n = \frac{1}{n+1} \binom{2n}{n},$$

so that its logarithmic capacity satisfies

$$\log_2 C_n = N - \frac{3}{2} \log_2 N + O(1).$$

Geometric neutrality is therefore compatible with an asymptotically extensive internal quantum state space.

Allowing a protected multiplicity to vary with the spacetime event produces a relational vector bundle. Its induced connection

$$A = Z^\dagger DZ$$

has nontrivial curvature only when the physical projector moves relative to the ambient comparison structure or when ambient curvature is present. A changing basis in one fixed subspace produces only pure gauge. The physical internal comparison group is correspondingly determined by holonomy,

$$G_{\text{int}}^{\text{dyn}} = \overline{\text{Hol}(\nabla^{\mathcal{M}})},$$

rather than by the multiplicity dimension alone. For a rank-two bundle, noncommuting transported curvature values generate the full $\mathfrak{su}(2)$ traceless holonomy algebra, whereas collinear curvature remains Abelian.

The $\mathbb{CP}^2$ dark-doublet model provides an explicit minimal example in which a moving two-dimensional relational subspace develops generic $u(2)$ local holonomy. It also demonstrates the distinction between local non-Abelianity and global topological generality: the corresponding bundle satisfies a restricted relation between its first and second Chern classes and cannot generate a nonzero $SU(2)$ instanton number on S^4 by simple pullback.

To obtain a universal internal interaction, the comparison transports of different sectors must be related. Requiring transport to commute with fusion leads to

$$U_\gamma^{(t)} C_{rs}^t = C_{rs}^t (U_\gamma^{(r)} \otimes U_\gamma^{(s)}),$$

and hence to a tensor-natural family of sector transports. Under the usual tensor-reconstruction hypotheses, the common internal comparison group is

$$G_{\text{int}} = \text{Aut}^\otimes(\omega),$$

and the different sector connections take the associated-representation form

$$A^{(r)} = \rho_{r*}(A).$$

In the Abelian case the same compatibility condition gives

$$q_t = q_r + q_s$$

for every allowed fusion channel. Charge additivity is thereby interpreted as preservation of the relational fusion structure under common internal comparison.

Finally, we have separated gauge reconstruction from gauge-field dynamics. A non-Abelian Berry or Wilczek–Zee connection can remain completely composite. A propagating Yang–Mills field requires connection liberation: the transverse infrared kernel must contain a massless pole with positive finite stiffness,

$$m_A^2 = 0, \quad 0 < K_A < \infty.$$

Only then can

$$K_A = g_{\text{eff}}^{-2}$$

be interpreted as the long-wavelength stiffness of an independently propagating internal comparison field.

The resulting logical chain is

$$\begin{aligned} \text{projective comparison} &\longrightarrow \text{relational multiplicity} \\ &\longrightarrow \text{noncommutative internal algebra} \\ &\longrightarrow \text{protected multiplicity bundle} \\ &\longrightarrow \text{curvature and holonomy} \\ &\longrightarrow \text{fusion-compatible common group} \\ &\longrightarrow \text{liberated gauge dynamics.} \end{aligned}$$

The first stages of this chain are consequences of representation theory, operator algebra, and vector-bundle geometry once the projective carrier is given. The later stages impose progressively stronger physical conditions. In particular, the present work does not claim a derivation of the Standard Model gauge group or its couplings. It identifies instead a candidate structural origin of internal

quantum degrees of freedom and makes explicit the additional dynamical tests required before those degrees of freedom become a physical gauge interaction.

Within the MS viewpoint, the distinction between geometry and internal structure can therefore be stated without introducing two unrelated primitive spaces. Geometry records the common representation by which localized carriers respond to spacetime comparison, whereas internal structure records the coherent multiplicity of relational realizations left invisible by that geometric action. Gauge comparison is the transport of this multiplicity in a way that is compatible with composition. This provides a relation-first route from projective quantum matter to internal gauge structure while preserving a sharp boundary between kinematic reconstruction and dynamical emergence.

A Spin- $\frac{1}{2}$ Multiplicities and Recoupling Conventions

This appendix fixes the spin-coupling conventions used in the main text and derives the multiplicity formulas needed for the minimal relational examples. All statements in this appendix concern ordinary finite-dimensional $SU(2)$ representation theory. Their role is to make explicit the algebraic input underlying the relational interpretation developed in the paper.

Let

$$V_{\frac{1}{2}} = \text{span}\{|\uparrow\rangle, |\downarrow\rangle\} \simeq \mathbb{C}^2$$

denote the fundamental spin- $\frac{1}{2}$ representation. For N carriers,

$$\mathcal{H}_N = \left(V_{\frac{1}{2}}\right)^{\otimes N}.$$

The collective generators are

$$J_{\text{tot}}^a = \sum_{i=1}^N J_i^a, \quad J_i^a = \frac{1}{2} \sigma_i^a,$$

where $\hbar = 1$ is used throughout the purely representation-theoretic calculations of the appendices.

The tensor product decomposes as

$$\mathcal{H}_N \simeq \bigoplus_j V_j \otimes M_j^{(N)}.$$

Let

$$m_j^{(N)} = \dim M_j^{(N)}$$

denote the multiplicity of the spin- j irreducible representation.

A convenient derivation begins with the number of product states carrying a given eigenvalue M of J_{tot}^3 . A state with $n_{\downarrow}$ down-spins has

$$M = \frac{1}{2} (N - 2n_{\downarrow}),$$

so that

$$n_{\downarrow} = \frac{N}{2} - M.$$

The dimension of the fixed- M subspace is therefore

$$d_M^{(N)} = \binom{N}{N/2 - M}.$$

Every copy of V_j contributes exactly one state to each magnetic subspace with

$$-j \leq M \leq j.$$

For $M \geq 0$ this gives

$$d_M^{(N)} = \sum_{j \geq M} m_j^{(N)}.$$

Taking the difference between adjacent values,

$$m_j^{(N)} = d_j^{(N)} - d_{j+1}^{(N)},$$

and using Eq. (A), one obtains

$$m_j^{(N)} = \binom{N}{N/2 - j} - \binom{N}{N/2 - j - 1}.$$

A binomial coefficient with a lower entry outside the interval $[0, N]$ is understood to vanish.

The same multiplicities satisfy the recursion

$$m_j^{(N+1)} = m_{j-\frac{1}{2}}^{(N)} + m_{j+\frac{1}{2}}^{(N)},$$

with the boundary convention that nonexistent spins have zero multiplicity. This follows directly from

$$V_j \otimes V_{\frac{1}{2}} \simeq V_{j+\frac{1}{2}} \oplus V_{j-\frac{1}{2}}.$$

Equation (A) provides a useful consistency check on Eq. (A).

For three carriers,

$$V_{\frac{1}{2}}^{\otimes 3} = (V_0 \oplus V_1) \otimes V_{\frac{1}{2}}$$

and hence

$$V_{\frac{1}{2}}^{\otimes 3} \simeq V_{\frac{3}{2}} \oplus V_{\frac{1}{2}} \oplus V_{\frac{1}{2}}.$$

Equivalently,

$$V_{\frac{1}{2}}^{\otimes 3} \simeq V_{\frac{3}{2}} \oplus (V_{\frac{1}{2}} \otimes \mathbb{C}^2).$$

The multiplicity factor $\mathbb{C}^2$ records the two distinct intermediate coupling channels

$$j_{12} = 0, \quad j_{12} = 1,$$

that can both lead to total spin $j = \frac{1}{2}$.

For four carriers,

$$V_{\frac{1}{2}}^{\otimes 4} \simeq V_2 \oplus 3V_1 \oplus 2V_0,$$

or, displaying the multiplicity factors explicitly,

$$V_{\frac{1}{2}}^{\otimes 4} \simeq V_2 \oplus (V_1 \otimes \mathbb{C}^3) \oplus (V_0 \otimes \mathbb{C}^2).$$

The invariant sector is therefore

$$\mathcal{I}_4 = \text{Inv}_{SU(2)} \left(V_{\frac{1}{2}}^{\otimes 4} \right) \simeq \mathbb{C}^2.$$

We now fix an explicit orthonormal basis for $\mathcal{I}_4$. For two spin- $\frac{1}{2}$ carriers define

$$|s\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

and

$$|t_+\rangle = |\uparrow\uparrow\rangle, \quad |t_0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), \quad |t_-\rangle = |\downarrow\downarrow\rangle.$$

Coupling the pairs (1, 2) and (3, 4) first, define

$$|0_L\rangle = |s_{12}\rangle|s_{34}\rangle$$

and

$$|1_L\rangle = \frac{1}{\sqrt{3}}(|t_{+,12}\rangle|t_{-,34}\rangle - |t_{0,12}\rangle|t_{0,34}\rangle + |t_{-,12}\rangle|t_{+,34}\rangle).$$

The relative sign of the middle term follows from the conventional Clebsch–Gordan coefficient for coupling two spin-one representations to total spin zero.

The two states satisfy

$$\langle 0_L|0_L\rangle = \langle 1_L|1_L\rangle = 1, \quad \langle 0_L|1_L\rangle = 0,$$

and

$$\mathbf{J}_{\text{tot}}|0_L\rangle = \mathbf{J}_{\text{tot}}|1_L\rangle = 0.$$

They therefore form an orthonormal basis of $\mathcal{I}_4$.

In the computational basis,

$$|0_L\rangle = \frac{1}{2}(|\uparrow\downarrow\uparrow\downarrow\rangle - |\uparrow\downarrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\downarrow\rangle + |\downarrow\uparrow\downarrow\uparrow\rangle),$$

whereas

$$\begin{aligned} |1_L\rangle = & \frac{1}{\sqrt{3}}(|\uparrow\uparrow\downarrow\downarrow\rangle + |\downarrow\downarrow\uparrow\uparrow\rangle) \\ & - \frac{1}{2\sqrt{3}}(|\uparrow\downarrow\uparrow\downarrow\rangle + |\uparrow\downarrow\downarrow\uparrow\rangle + |\downarrow\uparrow\uparrow\downarrow\rangle + |\downarrow\uparrow\downarrow\uparrow\rangle). \end{aligned}$$

Alternative pairing schemes define different bases of the same invariant space. Consider, for example,

$$|s_{13}s_{24}\rangle$$

and

$$|s_{14}s_{23}\rangle.$$

With the singlet orientation convention of Eq. (A), direct expansion gives

$$|s_{13}s_{24}\rangle = \frac{1}{2}|0_L\rangle + \frac{\sqrt{3}}{2}|1_L\rangle,$$

and

$$|s_{14}s_{23}\rangle = -\frac{1}{2}|0_L\rangle + \frac{\sqrt{3}}{2}|1_L\rangle.$$

In particular,

$$|\langle s_{12}s_{34}|s_{14}s_{23}\rangle| = \frac{1}{2}.$$

Different singlet-pairing patterns are therefore nonorthogonal states rather than mutually exclusive classical configurations.

For the recoupling from the (12)(34) basis to the (23)(14) basis, choose

$$|0'_L\rangle = |s_{14}s_{23}\rangle = -\frac{1}{2}|0_L\rangle + \frac{\sqrt{3}}{2}|1_L\rangle.$$

An orthonormal complementary singlet may be chosen as

$$|1'_L\rangle = \frac{\sqrt{3}}{2}|0_L\rangle + \frac{1}{2}|1_L\rangle.$$

Then

$$\begin{pmatrix} |0'_L\rangle \\ |1'_L\rangle \end{pmatrix} = F \begin{pmatrix} |0_L\rangle \\ |1_L\rangle \end{pmatrix}, \quad F = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}.$$

The matrix satisfies

$$F^\dagger F = I, \quad F^2 = I, \quad \det F = -1.$$

Different overall phases of the recoupled basis vectors modify individual signs in F but do not alter any projector or commutator used in the main text.

The dimension of the total-spin-zero sector for a general even number

$$N = 2n$$

of carriers follows from Eq. (A):

$$m_0^{(2n)} = \binom{2n}{n} - \binom{2n}{n-1}.$$

Using

$$\binom{2n}{n-1} = \frac{n}{n+1} \binom{2n}{n},$$

one finds

$$m_0^{(2n)} = \frac{1}{n+1} \binom{2n}{n} = C_n,$$

the n th Catalan number.

Stirling's expansion,

$$n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \left[1 + O\left(\frac{1}{n}\right)\right],$$

gives

$$\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}},$$

and therefore

$$C_n \sim \frac{4^n}{\sqrt{\pi} n^{3/2}}.$$

Since $N = 2n$,

$$\log_2 C_n = N - \frac{3}{2} \log_2 N + O(1),$$

which is the asymptotic capacity formula used in Section 4.

The multiplicity decomposition may also be viewed through the double-action structure of collective $SU(2)$ rotations and permutation operators. The collective rotation

$$U_{\frac{1}{2}}(g)^{\otimes N}$$

commutes with the natural permutation representation of the symmetric group S_N . Consequently, operators generated by permutations act nontrivially on the multiplicity factors while preserving the collective geometric representation. For the spin- $\frac{1}{2}$ case this centralizer description is equivalent, after restricting the allowed Young diagrams to at most two rows, to the multiplicity structure used throughout the paper [?, ?].

The interpretation adopted in the main text does not require the permutation group itself to be a physical internal gauge group. Its role here is to make explicit that a nontrivial operator algebra already exists in the commutant of the collective rotational action. The next appendix shows that the same commutant algebra can be generated directly by pair-neutrality observables with a relational interpretation.

B Pair Projectors and the Relational Qubit Algebra

We now derive the explicit pair-projector matrices used in Section 3 and show directly that rotationally invariant pair relations generate the full operator algebra on the four-carrier singlet multiplicity.

For two spin- $\frac{1}{2}$ carriers i and j , the total pair spin satisfies

$$(\mathbf{J}_i + \mathbf{J}_j)^2 = \mathbf{J}_i^2 + \mathbf{J}_j^2 + 2\mathbf{J}_i \cdot \mathbf{J}_j.$$

Since

$$\mathbf{J}_i^2 = \mathbf{J}_j^2 = \frac{3}{4}I,$$

the scalar product has eigenvalue

$$\mathbf{J}_i \cdot \mathbf{J}_j = -\frac{3}{4}$$

in the pair singlet and

$$\mathbf{J}_i \cdot \mathbf{J}_j = \frac{1}{4}$$

in the pair triplet. The singlet projector is therefore

$$P_{ij}^{(0)} = \frac{1}{4}I - \mathbf{J}_i \cdot \mathbf{J}_j.$$

Using

$$\mathbf{J}_i = \frac{1}{2}\boldsymbol{\sigma}_i,$$

this becomes

$$P_{ij}^{(0)} = \frac{1}{4}(I - \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j).$$

It satisfies

$$\left(P_{ij}^{(0)}\right)^2 = P_{ij}^{(0)}, \quad \left(P_{ij}^{(0)}\right)^\dagger = P_{ij}^{(0)}.$$

Because

$$\mathbf{J}_i \cdot \mathbf{J}_j$$

is a rotational scalar,

$$\left[P_{ij}^{(0)}, J_{\text{tot}}^a\right] = 0.$$

Every pair-singlet projector therefore preserves each total-spin sector and, in particular, restricts to an operator on $\mathcal{I}_4$.

In the logical basis

$$\{|0_L\rangle, |1_L\rangle\}$$

defined in Appendix A, the pair (1, 2) is a singlet in $|0_L\rangle$ and a triplet in $|1_L\rangle$. Hence

$$P_{12}^{(0)}|_{\mathcal{I}_4} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

By the same argument,

$$P_{34}^{(0)}|_{\mathcal{I}_4} = P_{12}^{(0)}|_{\mathcal{I}_4}.$$

For the pair (2, 3), the eigenstate with eigenvalue one is the recoupled singlet state

$$|s_{14}s_{23}\rangle = -\frac{1}{2}|0_L\rangle + \frac{\sqrt{3}}{2}|1_L\rangle.$$

Therefore

$$P_{23}^{(0)}|_{\mathcal{I}_4} = |s_{14}s_{23}\rangle\langle s_{14}s_{23}|,$$

which gives

$$P_{23}^{(0)}|_{\mathcal{I}_4} = \begin{pmatrix} \frac{1}{4} & -\frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}.$$

The same matrix is obtained for the pair (1, 4):

$$P_{14}^{(0)}|_{\mathcal{I}_4} = \begin{pmatrix} \frac{1}{4} & -\frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}.$$

For the alternative pairings (1, 3) and (2, 4), one instead uses Eq. (A). The result is

$$P_{13}^{(0)}|_{\mathcal{I}_4} = P_{24}^{(0)}|_{\mathcal{I}_4} = \begin{pmatrix} \frac{1}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}.$$

Thus all six pair-neutrality projectors reduce to only three distinct logical directions:

$$P_{12}^{(0)} = P_{34}^{(0)},$$

$$P_{23}^{(0)} = P_{14}^{(0)},$$

and

$$P_{13}^{(0)} = P_{24}^{(0)}$$

when restricted to $\mathcal{I}_4$.

The three projectors obey

$$P_{12}^{(0)} + P_{13}^{(0)} + P_{14}^{(0)} = \frac{3}{2}I_{\mathcal{I}_4}.$$

This relation is a consequence of total-spin neutrality. Indeed,

$$\mathbf{J}_{\text{tot}}^2 = \sum_{i=1}^4 \mathbf{J}_i^2 + 2 \sum_{i<j} \mathbf{J}_i \cdot \mathbf{J}_j = 0$$

on $\mathcal{I}_4$. Since

$$\sum_i \mathbf{J}_i^2 = 3I,$$

one obtains

$$\sum_{i<j} \mathbf{J}_i \cdot \mathbf{J}_j = -\frac{3}{2}I,$$

which is equivalent to the corresponding relation among singlet projectors.

The noncommutativity relevant to the main text follows immediately:

$$[P_{12}^{(0)}, P_{23}^{(0)}] = \frac{\sqrt{3}}{4} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Defining the standard logical Pauli matrices

$$X_L = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y_L = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z_L = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

one may write

$$P_{12}^{(0)} = \frac{1}{2}(I + Z_L),$$

$$P_{23}^{(0)} = \frac{1}{2}I - \frac{1}{4}Z_L - \frac{\sqrt{3}}{4}X_L,$$

and

$$P_{13}^{(0)} = \frac{1}{2}I - \frac{1}{4}Z_L + \frac{\sqrt{3}}{4}X_L.$$

Consequently,

$$[P_{12}^{(0)}, P_{23}^{(0)}] = -\frac{i\sqrt{3}}{4}Y_L.$$

The three pair-neutrality axes therefore lie in a common great circle of the logical Bloch sphere and are separated by angles of 120° in their traceless Bloch vectors. For example, the traceless parts are proportional to

$$\mathbf{n}_{12} = (0, 0, 1),$$

$$\mathbf{n}_{23} = \left(-\frac{\sqrt{3}}{2}, 0, -\frac{1}{2}\right),$$

and

$$\mathbf{n}_{13} = \left(\frac{\sqrt{3}}{2}, 0, -\frac{1}{2}\right),$$

which satisfy

$$\mathbf{n}_{12} + \mathbf{n}_{23} + \mathbf{n}_{13} = 0.$$

Thus different choices of neutral pairing correspond to different, incompatible logical measurement directions rather than to orthogonal classical labels.

The complete matrix algebra follows directly. From Eq. (B),

$$Z_L = 2P_{12}^{(0)} - I.$$

Using Eq. (B),

$$X_L = -\frac{4}{\sqrt{3}} \left(P_{23}^{(0)} - \frac{1}{2}I + \frac{1}{4}Z_L \right).$$

Finally,

$$Y_L = \frac{1}{2i}[Z_L, X_L].$$

The unital $*$ -algebra generated by $P_{12}^{(0)}$ and $P_{23}^{(0)}$ therefore contains

$$I, \quad X_L, \quad Y_L, \quad Z_L.$$

Since these form a complex linear basis of $M_2(\mathbb{C})$,

$$\text{Alg}^*(P_{12}^{(0)}, P_{23}^{(0)}) = M_2(\mathbb{C}).$$

This calculation also shows that no fundamental one-body operator is required to manipulate the logical qubit. A general Hermitian logical operator

$$H_L = aI + bX_L + cY_L + dZ_L$$

can be expressed as a polynomial in rotationally invariant pair relations. The logical state can therefore undergo arbitrary unitary evolution while remaining inside the total-spin-zero geometric sector.

The relation to exchange operators is immediate. Let

$$\Pi_{ij}$$

denote the operator that exchanges two spin- $\frac{1}{2}$ carriers. It acts as -1 on their antisymmetric singlet and $+1$ on their symmetric triplet, so

$$\Pi_{ij} = I - 2P_{ij}^{(0)}.$$

Equivalently,

$$\Pi_{ij} = \frac{1}{2} (I + \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j).$$

Thus the pair-neutrality algebra and the exchange algebra generate the same centralizer structure.

It is also useful to introduce

$$e_i = 2P_{i,i+1}^{(0)}.$$

Since $P_{i,i+1}^{(0)}$ is a projector,

$$e_i^2 = 2e_i.$$

For three adjacent spin- $\frac{1}{2}$ carriers, singlet recoupling gives

$$P_{i,i+1}^{(0)} P_{i+1,i+2}^{(0)} P_{i,i+1}^{(0)} = \frac{1}{4} P_{i,i+1}^{(0)}.$$

Multiplying by the appropriate factors of two yields

$$e_i e_{i+1} e_i = e_i,$$

and similarly

$$e_i e_{i-1} e_i = e_i.$$

Operators acting on disjoint pairs commute:

$$[e_i, e_j] = 0, \quad |i - j| > 1.$$

Equations (B), (B), and (B) are the Temperley–Lieb relations with loop parameter

$$\delta = 2.$$

For completeness, the coefficient in Eq. (B) can be seen directly from the three-spin recoupling amplitude. If carriers (1, 2) are initially in a singlet, the probability amplitude for (2, 3) to be found in a singlet and then for (1, 2) to return to its original singlet channel is fixed by the spin- $\frac{1}{2}$ recoupling coefficient. The corresponding squared overlap produces the factor

$$\frac{1}{4}.$$

The Temperley–Lieb relation is therefore another algebraic expression of the nonorthogonality of different local pairing decompositions.

The appearance of the Temperley–Lieb algebra should not be interpreted as an additional microscopic postulate of the MS theory. It follows automatically from the tensor category of ordinary spin- $\frac{1}{2}$ carriers. Its relevance here is that its generators possess a direct relational interpretation:

$$e_i = 2P_{i,i+1}^{(0)}$$

tests or energetically weights the neutral relation between neighboring projective carriers while commuting with collective rotation.

There is an important distinction between this noncommutativity and a scalar central quantum phase. For two nonorthogonal rank-one projectors P and Q on a two-dimensional real subspace, let

$$t = \text{Tr}(PQ), \quad 0 < t < 1.$$

A direct calculation gives

$$[P, Q]^2 = -t(1 - t)I.$$

One may therefore define

$$J_{P,Q} = \frac{[P, Q]}{\sqrt{t(1 - t)}},$$

for which

$$J_{P,Q}^2 = -I.$$

However, $J_{P,Q}$ is generated by the relative orientation of two noncommuting projectors. It is an internal oriented-plane or symplectic generator; it is not a central scalar complex unit acting identically on the full physical algebra. In particular, it should not be identified with the universal scalar i of the Hilbert-space representation.

For the pair

$$P = P_{12}^{(0)}, \quad Q = P_{23}^{(0)}$$

on $\mathcal{I}_4$,

$$t = \text{Tr}(PQ) = \frac{1}{4},$$

and hence

$$t(1-t) = \frac{3}{16}.$$

Equation (B) then gives

$$J_{P,Q} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -iY_L,$$

which indeed satisfies

$$J_{P,Q}^2 = -I.$$

This provides a useful internal orientation on the logical qubit, but its origin remains relational rather than central.

The finite-dimensional results of this appendix can therefore be summarized without invoking any gauge interpretation. Collective rotation leaves a multiplicity space unresolved; rotationally invariant pair projectors act noncommutatively on that space; two suitable pair projectors generate the complete logical matrix algebra; and the same generators realize the Temperley–Lieb centralizer structure. The identification of these multiplicity algebras with candidate internal quantum degrees of freedom is the physical step developed in the main text.

C Projected One-Body Operators and Second-Order Relational Response

This appendix derives the geometric-blindness relations used in Section 4 and makes explicit the leading effective response induced inside a total-spin-neutral relational sector by local one-body vector perturbations.

Let

$$\mathcal{H}_N = \left(V_{\frac{1}{2}}\right)^{\otimes N}$$

and let

$$\mathcal{I}_N = \text{Inv}_{SU(2)}(\mathcal{H}_N)$$

denote the total-spin-zero subspace for even N . The orthogonal projector onto this space will be denoted by

$$P = P_{\text{inv}}, \quad Q = I - P.$$

The collective rotation is generated by

$$J_{\text{tot}}^a = \sum_{i=1}^N J_i^a.$$

For every $g \in SU(2)$,

$$U_N(g)P = PU_N(g) = P,$$

when the equality is understood as an operator relation on the invariant subspace. A local spin operator transforms as a rank-one rotational tensor,

$$U_N(g)J_i^a U_N(g)^{-1} = R(g)^a_b J_i^b.$$

Projecting this equation onto $\mathcal{I}_N$ gives

$$PJ_i^a P = R(g)^a_b PJ_i^b P \quad \forall g \in SU(2).$$

The three operators

$$A_i^a := PJ_i^a P$$

would therefore define an invariant vector under the ordinary three-dimensional rotation representation. No such nonzero invariant vector exists. Hence

$$PJ_i^a P = 0.$$

This may equivalently be viewed as the rank-one case of the Wigner–Eckart theorem: a vector operator cannot have a matrix element between two total-spin-zero states.

Lemma C.1 (Vanishing of projected vector operators). Let P project onto any direct sum of trivial $SU(2)$ representations. If V^a transforms as a rotational vector, then

$$PV^a P = 0.$$

In particular,

$$PJ_i^a P = 0$$

for every constituent carrier i .

The statement is operator valued and is stronger than

$$\langle J_i^a \rangle = 0.$$

It implies

$$\langle \phi | J_i^a | \psi \rangle = 0$$

for every pair

$$|\phi\rangle, |\psi\rangle \in \mathcal{I}_N.$$

Thus no local vector operator can distinguish, mix, or otherwise act directly within the neutral relational state space at first order.

Consider now a weak one-body geometric perturbation

$$V = \sum_{i=1}^N \Omega_i^a J_i^a = \sum_{i=1}^N \mathbf{\Omega}_i \cdot \mathbf{J}_i,$$

where $\mathbf{\Omega}_i$ may represent a local rotational comparison field, an effective Zeeman-like source, or any other perturbation transforming as a geometric vector at the i th carrier. Equation (C) immediately gives

$$PVP = 0.$$

Suppose the unperturbed Hamiltonian H_0 possesses the protected relational sector $P\mathcal{H}_N$ with energy E_0 ,

$$H_0 P = E_0 P,$$

and assume that the complementary sector is separated from E_0 so that

$$Q(H_0 - E_0)Q$$

is invertible on $Q\mathcal{H}_N$. The standard Feshbach or Schrieffer–Wolff expansion gives the effective Hamiltonian on the protected sector,

$$H_{\text{eff}} = E_0 P + PVP - PVQ[Q(H_0 - E_0)Q]^{-1}QVP + O(V^3).$$

Since

$$PVP = 0,$$

the leading state-dependent correction is

$$H_{\text{eff}}^{(2)} = -PVQ[Q(H_0 - E_0)Q]^{-1}QVP.$$

If the relevant intermediate states can be approximated by a common positive gap

$$\Delta > 0,$$

so that

$$Q(H_0 - E_0)Q \simeq \Delta Q,$$

then

$$H_{\text{eff}}^{(2)} \simeq -\frac{1}{\Delta}PVQVP.$$

Using

$$Q = I - P$$

and

$$PVP = 0,$$

one obtains

$$PVQVP = PV^2P,$$

and therefore

$$H_{\text{eff}}^{(2)} \simeq -\frac{1}{\Delta}PV^2P.$$

The square of the perturbation is

$$V^2 = \sum_{i,j} \Omega_i^a \Omega_j^b J_i^a J_j^b.$$

It remains to determine the projection of the rank-two tensor $J_i^a J_j^b$ onto the rotationally invariant sector.

Inside a singlet sector, the operator

$$PJ_i^a J_j^b P$$

must itself be invariant under simultaneous rotation of the two vector indices. The only invariant rank-two tensor is δ^{ab} . Hence

$$PJ_i^a J_j^b P = \delta^{ab} X_{ij}$$

for some operator X_{ij} acting within $\mathcal{I}_N$. Contracting the indices gives

$$P(\mathbf{J}_i \cdot \mathbf{J}_j)P = 3X_{ij}.$$

Therefore

$$PJ_i^a J_j^b P = \frac{\delta^{ab}}{3} P(\mathbf{J}_i \cdot \mathbf{J}_j)P.$$

Lemma C.2 (Scalar reduction of two-vector response). For any pair of constituent vector operators acting between total-spin-zero states,

$$PJ_i^a J_j^b P = \frac{\delta^{ab}}{3} P(\mathbf{J}_i \cdot \mathbf{J}_j)P.$$

The antisymmetric vector component and the symmetric-traceless spin-two component have vanishing matrix elements between singlet states.

Substitution into Eq. (C) gives

$$H_{\text{eff}}^{(2)} \simeq -\frac{1}{3\Delta} \sum_{i,j} (\boldsymbol{\Omega}_i \cdot \boldsymbol{\Omega}_j) P(\mathbf{J}_i \cdot \mathbf{J}_j)P.$$

The local geometric perturbation has therefore been converted into an effective Hamiltonian built entirely from rotational scalars.

The terms with $i = j$ contribute only a common energy shift. Since

$$\mathbf{J}_i^2 = \frac{3}{4}I,$$

their contribution is

$$H_{\text{self}}^{(2)} = -\frac{1}{4\Delta} \sum_i |\boldsymbol{\Omega}_i|^2 P.$$

Nontrivial logical dynamics is generated by the cross terms

$$i \neq j.$$

The four-carrier relational qubit provides a useful explicit example. Using

$$P_{ij}^{(0)} = \frac{1}{4}I - \mathbf{J}_i \cdot \mathbf{J}_j,$$

the pair scalar operators restricted to $\mathcal{I}_4$ are

$$P \mathbf{J}_1 \cdot \mathbf{J}_2 P = -\frac{1}{4}I - \frac{1}{2}Z_L,$$

$$P \mathbf{J}_2 \cdot \mathbf{J}_3 P = -\frac{1}{4}I + \frac{1}{4}Z_L + \frac{\sqrt{3}}{4}X_L,$$

and

$$P \mathbf{J}_1 \cdot \mathbf{J}_3 P = -\frac{1}{4}I + \frac{1}{4}Z_L - \frac{\sqrt{3}}{4}X_L.$$

The remaining pairs satisfy

$$P \mathbf{J}_3 \cdot \mathbf{J}_4 P = P \mathbf{J}_1 \cdot \mathbf{J}_2 P,$$

$$P \mathbf{J}_1 \cdot \mathbf{J}_4 P = P \mathbf{J}_2 \cdot \mathbf{J}_3 P,$$

and

$$P \mathbf{J}_2 \cdot \mathbf{J}_4 P = P \mathbf{J}_1 \cdot \mathbf{J}_3 P.$$

Define the three geometric overlap combinations

$$C_{12} = \boldsymbol{\Omega}_1 \cdot \boldsymbol{\Omega}_2 + \boldsymbol{\Omega}_3 \cdot \boldsymbol{\Omega}_4,$$

$$C_{23} = \boldsymbol{\Omega}_2 \cdot \boldsymbol{\Omega}_3 + \boldsymbol{\Omega}_1 \cdot \boldsymbol{\Omega}_4,$$

and

$$C_{13} = \boldsymbol{\Omega}_1 \cdot \boldsymbol{\Omega}_3 + \boldsymbol{\Omega}_2 \cdot \boldsymbol{\Omega}_4.$$

After separating the state-independent part, the logical contribution to Eq. (C) is

$$H_L^{(2)} = h_X X_L + h_Z Z_L + \text{constant},$$

with

$$h_X = -\frac{\sqrt{3}}{6\Delta} (C_{23} - C_{13})$$

and

$$h_Z = \frac{1}{6\Delta} (2C_{12} - C_{23} - C_{13}).$$

Thus a perturbation constructed entirely from local geometric vectors can produce nontrivial internal qubit dynamics at second order even though every first-order projected vector operator vanishes.

For real static perturbations in the common-gap approximation, Eq. (C) lies in the X_L – Z_L plane. The missing logical generator Y_L can nevertheless arise from noncommuting sequential effective operations. If

$$H_A = a_X X_L + a_Z Z_L$$

and

$$H_B = b_X X_L + b_Z Z_L,$$

then

$$[H_A, H_B] = 2i(a_X b_Z - a_Z b_X) Y_L.$$

Consequently, time-dependent or piecewise comparison perturbations can generate the full logical Lie algebra even when each instantaneous second-order Hamiltonian is real in the chosen logical basis.

The simple common-gap approximation is not required for the geometric blindness result. If the intermediate spectrum contains different energies, write

$$Q = \sum_{\lambda} Q_{\lambda}$$

with

$$H_0 Q_{\lambda} = E_{\lambda} Q_{\lambda}.$$

Then

$$H_{\text{eff}}^{(2)} = - \sum_{\lambda} \frac{PVQ_{\lambda}VP}{E_{\lambda} - E_0}.$$

Different intermediate rotational sectors may therefore weight different relational channels differently. Such spectral structure can generate a richer effective multiplicity Hamiltonian than Eq. (C).

The derivation also shows why a protective gap and internal dynamics should not be conflated. A positive gap

$$\Delta_{\text{rel}} > 0$$

suppresses real leakage out of the protected relational sector, while virtual excursions across the same gap generate effective internal operators of order

$$\frac{V^2}{\Delta_{\text{rel}}}.$$

Protection and responsiveness are therefore compatible: the relational sector can remain spectrally isolated while still acquiring controlled effective dynamics.

More generally, if the perturbation contains a scalar part

$$V_{\text{scalar}} = \sum_{i < j} \lambda_{ij} \mathbf{J}_i \cdot \mathbf{J}_j,$$

then

$$PV_{\text{scalar}}P \neq 0$$

in general, and relational information can be manipulated directly at first order. The first-order blindness theorem applies specifically to geometric operators transforming nontrivially under the collective rotational action, such as one-body vectors. Rotationally invariant relational operators belong to the commutant and may act directly inside the multiplicity sector.

The logical distinction is therefore

$$\text{geometric vector probe} \implies PVP = 0,$$

whereas

$$\text{relational scalar probe} \implies PVP \text{ may be nonzero.}$$

This provides an operator-theoretic meaning to the separation between geometric and internal comparison used throughout the main text.

Finally, nothing in the argument implies absolute isolation of the multiplicity sector from space-time geometry. The result is relative to a specified collective rotational comparison algebra and to perturbations transforming as one-body geometric vectors. Higher-rank geometric operators, inhomogeneous environments, curvature couplings, or microscopic interactions may couple directly to other components of the relational algebra. The physical content of the result is narrower: geometry-neutral multiplicity information is not carried by one-body rotational polarization, and its leading response to such polarization is mediated by relational composite operators.

D Projector Connections and the Relational Gauss Identity

This appendix derives the projector and frame formulas used in Section 5. The discussion is standard differential geometry of Hermitian subbundles and universal connections [?, ?, ?, ?]. We include the derivation to make explicit which part of the internal curvature is due to the ambient comparison connection and which part is generated by the motion of the protected relational subspace itself.

Let

$$\mathcal{R} \longrightarrow M$$

be a rank- N Hermitian vector bundle with unitary connection

$$D.$$

Its curvature is

$$F_D = D^2.$$

Let

$$P = P^\dagger = P^2$$

be a smooth rank- m orthogonal projector on $\mathcal{R}$. Its image defines a Hermitian subbundle

$$\mathcal{M} = \text{im} P \longrightarrow M.$$

Write

$$Q = I - P.$$

The connection induced on $\mathcal{M}$ is

$$\nabla^{\mathcal{M}} = PD$$

when acting on sections satisfying

$$P\psi = \psi.$$

Its curvature is

$$F_{\mathcal{M}} = \left(\nabla^{\mathcal{M}} \right)^2.$$

For a section

$$\psi = P\psi$$

of $\mathcal{M}$,

$$\nabla^{\mathcal{M}}\psi = PD\psi.$$

Applying the projected derivative once more gives

$$F_{\mathcal{M}}\psi = PD(PD\psi).$$

Using

$$D(P\eta) = (DP)\eta + PD\eta$$

for an arbitrary bundle-valued form η , one obtains

$$F_{\mathcal{M}}\psi = P(DP) \wedge D\psi + PD^2\psi.$$

Since

$$\psi = P\psi,$$

its ambient derivative decomposes as

$$D\psi = PD\psi + QD\psi.$$

Differentiating

$$P\psi = \psi$$

also gives

$$QD\psi = Q(DP)\psi.$$

Moreover,

$$P(DP)P = 0,$$

which follows by differentiating

$$P^2 = P :$$

$$(DP)P + P(DP) = DP$$

and multiplying by P on both sides.

The tangential part of $D\psi$ therefore drops out of the first term in Eq. (D), leaving

$$F_{\mathcal{M}}\psi = P(DP) \wedge Q(DP)P\psi + PF_D P\psi.$$

Since

$$P(DP) = P(DP)Q$$

and

$$(DP)P = Q(DP)P,$$

this can be written

$$F_{\mathcal{M}} = PF_D P + P(DP \wedge DP)P.$$

Theorem D.1 (Projector Gauss identity). For a Hermitian subbundle selected by an orthogonal projector P inside an ambient Hermitian bundle with unitary connection D ,

$$F_{\mathcal{M}} = PF_D P + P(DP \wedge DP)P.$$

The first term is inherited from ambient comparison curvature; the second is generated by the extrinsic motion of the physical subspace.

This formula is gauge covariant and depends only on the physical projector. No choice of basis inside $\mathcal{M}$ is required.

To connect it with the local matrix formulas used in the main text, choose an orthonormal frame

$$Z = (|m_1\rangle, \dots, |m_m\rangle)$$

for $\mathcal{M}$. It may be viewed as a bundle map

$$Z : \mathbb{C}^m \longrightarrow \mathcal{R}$$

satisfying

$$Z^\dagger Z = I_m, \quad ZZ^\dagger = P.$$

Define the induced connection matrix by

$$A = Z^\dagger DZ.$$

Metric compatibility of D implies

$$A^\dagger = -A,$$

so

$$A \in \Omega^1(M, \mathfrak{u}(m)).$$

The component of DZ orthogonal to the protected subspace is

$$K = QDZ.$$

Because

$$PDZ = ZZ^\dagger DZ = ZA,$$

the frame derivative decomposes as

$$DZ = ZA + K.$$

This is the bundle analogue of the usual tangential–normal decomposition in submanifold geometry.

The frame transformation

$$Z \mapsto ZU, \quad U : M \longrightarrow U(m),$$

leaves

$$P = ZZ^\dagger$$

unchanged. The induced fields transform as

$$A \mapsto U^{-1}AU + U^{-1}dU$$

and

$$K \mapsto KU.$$

Thus A has the transformation law of a connection, while K transforms homogeneously.

The decomposition also follows directly at the level of physical states. A section of $\mathcal{M}$ can be written

$$|\Psi\rangle = Z\psi,$$

where

$$\psi : M \longrightarrow \mathbb{C}^m$$

contains the coordinates of the relational state in the chosen frame. Then

$$D|\Psi\rangle = (DZ)\psi + Z d\psi.$$

Using Eq. (D),

$$D|\Psi\rangle = Z(d + A)\psi + K\psi.$$

The first term describes internal comparison within the protected multiplicity space. The second describes the change of its embedding inside the larger comparison space.

We now derive the curvature identity in frame form. The curvature of A is

$$F_A = dA + A \wedge A.$$

Apply D to Eq. (D):

$$D^2Z = D(ZA + K).$$

Since

$$D^2Z = F_DZ,$$

one finds

$$F_DZ = ZF_A + K \wedge A + DK.$$

Multiplication by $Z^\dagger$ gives

$$Z^\dagger F_DZ = F_A + Z^\dagger DK,$$

because

$$Z^\dagger K = 0.$$

Differentiate this orthogonality relation:

$$D(Z^\dagger K) = 0.$$

Metric compatibility gives

$$(DZ)^\dagger \wedge K + Z^\dagger DK = 0.$$

Using

$$DZ = ZA + K$$

and

$$Z^\dagger K = 0,$$

one obtains

$$Z^\dagger DK = -K^\dagger \wedge K.$$

Substitution into Eq. (D) yields

$$F_A = Z^\dagger F_DZ + K^\dagger \wedge K.$$

Theorem D.2 (Frame Gauss identity). With the anti-Hermitian convention

$$A = Z^\dagger DZ, \quad F_A = dA + A \wedge A,$$

the induced curvature satisfies

$$F_A = Z^\dagger F_DZ + K^\dagger \wedge K, \quad K = (I - P)DZ.$$

Changing to a Hermitian gauge-potential convention moves the corresponding factors of i and may alter the displayed sign of the quadratic term, but not the invariant content of the relation.

The equivalence of Eqs. (D) and (D) can be checked directly. Differentiating

$$P = ZZ^\dagger$$

covariantly and using

$$DZ = ZA + K$$

gives

$$DP = KZ^\dagger + ZK^\dagger.$$

The internal connection A cancels because it only rotates the basis inside the same physical subspace. Consequently,

$$Z^\dagger(DP \wedge DP)Z = K^\dagger \wedge K.$$

This makes the geometric meaning of the extrinsic curvature term explicit: it depends on motion of the projector, not on a change of internal frame.

An immediate corollary is the fixed-subspace result used in the main text.

Corollary D.3 (No curvature from a basis change alone). Suppose the ambient bundle is locally flat,

$$F_D = 0,$$

and the physical projector is covariantly constant,

$$DP = 0.$$

Then

$$F_A = 0.$$

In a trivialization with $D = d$ and constant projector $P = P_0$, every local orthonormal frame can be written

$$Z = Z_0 U(x),$$

and

$$A = U^{-1} dU.$$

The connection is therefore pure gauge.

This is the precise sense in which a multiplicity-space basis freedom does not by itself constitute a physical gauge field. Nonzero induced curvature requires either

$$F_D \neq 0$$

or

$$DP \neq 0.$$

The flat-ambient case is particularly transparent:

$$F_D = 0 \implies F_A = K^\dagger \wedge K.$$

All internal curvature is then generated by the extrinsic response of the protected relational subspace.

In local spacetime coordinates, write

$$K = K_\mu dx^\mu.$$

Define the matrix-valued relational quantum-geometric tensor

$$Q_{\mu\nu} = K_\mu^\dagger K_\nu.$$

Since

$$Q_{\nu\mu} = Q_{\mu\nu}^\dagger,$$

the diagonal components are positive semidefinite:

$$Q_{\mu\mu} = K_\mu^\dagger K_\mu \geq 0.$$

For a flat ambient connection, Eq. (D) gives

$$F_{\mu\nu} = K_\mu^\dagger K_\nu - K_\nu^\dagger K_\mu = Q_{\mu\nu} - Q_{\nu\mu}.$$

The curvature is therefore the anti-Hermitian antisymmetric component of the same tensor whose symmetric Hermitian part measures the change of the protected subspace.

A scalar relational metric can be defined by

$$g_{\mu\nu}^{\text{rel}} = \frac{1}{2} \text{Tr}(Q_{\mu\nu} + Q_{\nu\mu}).$$

In particular,

$$g_{\mu\mu}^{\text{rel}} = \text{Tr}(K_\mu^\dagger K_\mu) = \|K_\mu\|_{\text{HS}}^2.$$

This scalar metric should not be identified with the spacetime metric reconstructed from the MS soldering form. It measures distinguishability of neighboring protected relational subspaces inside the ambient comparison Hilbert space.

The curvature satisfies a simple norm bound. By the triangle inequality,

$$\|F_{\mu\nu}\|_{\text{HS}} \leq \|K_\mu^\dagger K_\nu\|_{\text{HS}} + \|K_\nu^\dagger K_\mu\|_{\text{HS}}.$$

Using submultiplicativity of the Hilbert–Schmidt norm,

$$\|K_\mu^\dagger K_\nu\|_{\text{HS}} \leq \|K_\mu\|_{\text{HS}} \|K_\nu\|_{\text{HS}},$$

one obtains

$$\|F_{\mu\nu}\|_{\text{HS}} \leq 2\|K_\mu\|_{\text{HS}} \|K_\nu\|_{\text{HS}}.$$

Hence

$$\|F_{\mu\nu}\|_{\text{HS}} \leq 2\sqrt{g_{\mu\mu}^{\text{rel}} g_{\nu\nu}^{\text{rel}}}.$$

Corollary D.4 (Curvature–susceptibility bound). For a protected multiplicity subspace embedded in a flat ambient comparison bundle,

$$\|F_{\mu\nu}\|_{\text{HS}} \leq 2\sqrt{g_{\mu\mu}^{\text{rel}} g_{\nu\nu}^{\text{rel}}}.$$

Nonzero induced curvature therefore requires nonzero response of the physical projector to comparison deformations in the corresponding directions.

The bound should not be interpreted as an equation determining gauge stiffness. The tensor

$$g_{\mu\nu}^{\text{rel}}$$

is a kinematic measure of subspace motion, whereas the coefficient of

$$\text{tr } F_{\mu\nu} F^{\mu\nu}$$

in an effective action is a dynamical response coefficient. A rapidly varying relational projector may generate large Berry curvature without producing an independently propagating gauge field.

The Grassmannian formulation follows immediately. In a local trivialization of an N -dimensional ambient comparison bundle, every rank- m orthogonal projector defines a point

$$P \in \text{Gr}(m, N).$$

A spacetime-dependent protected subspace is therefore a map

$$P : M \longrightarrow \text{Gr}(m, N).$$

The bundle $\mathcal{M}$ is the pullback of the tautological bundle,

$$\mathcal{M} = P^* \mathcal{T}_m,$$

where

$$\mathcal{T}_m \longrightarrow \text{Gr}(m, N)$$

denotes the universal rank- m subbundle.

The connection

$$A = Z^\dagger dZ$$

in the flat ambient case is correspondingly the pullback of the universal connection on the tautological bundle. Its curvature is determined by the Grassmannian embedding through

$$F = P(dP \wedge dP)P.$$

This formulation makes it clear why the local frame Z is not the fundamental field: different frames related by

$$Z \rightarrow ZU$$

represent the same point of the Grassmannian and hence the same protected relational subspace.

The projector formulation also separates topological and dynamical questions. Characteristic classes of $\mathcal{M}$ can be computed from invariant polynomials of the curvature, for example

$$c_1(\mathcal{M}) = \left[\frac{i}{2\pi} \text{Tr} F_A \right],$$

and

$$\text{ch}_2(\mathcal{M}) = \left[-\frac{1}{8\pi^2} \text{Tr}(F_A \wedge F_A) \right]$$

with the present anti-Hermitian convention. Nonzero characteristic classes obstruct a global frame and force nontrivial patching of relational comparison states.

Topological nontriviality does not, however, determine the low-energy gauge kinetic term. The characteristic classes depend on the global bundle and connection topology, whereas a Yang–Mills stiffness is encoded in the dynamical effective action. Thus

$$c_k(\mathcal{M}) \neq 0$$

does not imply

$$g^{-2} > 0,$$

just as

$$F_A \neq 0$$

does not imply that A possesses an independent propagating degree of freedom.

The same formalism extends immediately to a nonflat ambient comparison connection. In that case,

$$F_A = Z^\dagger F_D Z + K^\dagger \wedge K.$$

The two terms have distinct interpretations. The first transfers ambient comparison curvature into the protected relational sector; the second is generated by its extrinsic embedding. Only their sum is intrinsic to the induced connection on $\mathcal{M}$.

This distinction can become important in an MS completion containing both geometric and non-geometric comparison structures. If the ambient bundle already carries a connection related to a more microscopic comparison dynamics, part of the effective internal curvature may be inherited rather than generated solely by the moving multiplicity projector. The projector formalism remains valid in either case and prevents the two contributions from being conflated.

Finally, the Gauss identity also clarifies the relation between protection and leakage. The operator

$$K_\mu = QD_\mu Z$$

measures the infinitesimal tendency of a protected state to leave the multiplicity subspace under ambient comparison. The same object determines the induced curvature quadratically:

$$F_{\mu\nu}^{\text{extr}} = K_\mu^\dagger K_\nu - K_\nu^\dagger K_\mu.$$

Thus a subspace that is perfectly rigid under all ambient comparison directions,

$$K_\mu = 0,$$

has no extrinsic Berry curvature. Nontrivial internal holonomy requires a controlled failure of covariant rigidity: the protected sector must remain spectrally well defined while its embedding changes from event to event.

The resulting hierarchy is

$$\text{protected relational subspace} \longrightarrow P(x) \longrightarrow K = (I - P)DZ \longrightarrow A = Z^\dagger DZ \longrightarrow F_A.$$

A frame change modifies A but not P , while physical subspace motion is encoded in DP or equivalently in K . This is the geometric distinction underlying the holonomy analysis of Section 6.